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--- Summary ---
: *** 951753
Coca Cola 2 ■
Simba chips 125 g – salt & vinegar
Yoghurt 6 pack – strawberry
All Gold tomato sauce
OMO washing powder 2 kg
TOTAL
CASH
CHANGE R24,95
R15,90
R21,90

R35,90
R87,95
R186,60
R200,00
R13,40
VAT 0% R0,00
VAT 15% R24,34
25/09/2024   16:10   Cashier : Belinda

1.2 Use the graph above to answer the questions that follow.
(2)

2.2 The Smith family is considering the following travel package to the Kruger National Park and it includes the following:

- Scheduled return flights from Johannesburg to the Kruger
- All road transfers between airport and activities
- 2 nights ' accommodation at Sudwala Lodge
- 4 nights ' accommodation at Sabie River Bush Lodge
- All breakfast and dinners included
- Elephant interaction where you and the little ones can feed, greet and walk hand -in trunk with the gentle giants
- Early morning and late afternoon off-road game drives that take place in an open 4×4 safari vehicle with a ranger and tracker team
- Child friendly facilities and kids activities
- Park and conservation fees

The cost for this package is from R 24 999 per person sharing and is only available during low season.

TAB LE 2: 10 FASTEST TIMES OF THE MEN'S 100 METER EVENT IN SOUTH AFRICA

	NAME AND SURNAME	Time (sec)	Date of birth	Place of birth
1	Akani Simbine	09,82	21 September 1993	Kempton Park
2	Benjamin Richardson	09,86	19 December 2003	Pretoria
3	Shaun Maswanganyi	09,87	01 February 2001	Soweto
4	Henricho Bruintjies	09,89	16 July 1993	Paarl
5	Retshidisitswe Mlenga	09,93	27 February 2000	Bloemfontein
6	Wayde van Niekerk	09,94	15 July 1992	Kraaifontein
6	Gift Leotlela	09,94	12 May 1998	Phuthaditjhaba
8	Thando Roto	A	26 September 1995	King Williams

Town
9 Simon Magakwe 09,98 14 May 1986 Itsoseng
10 Emile Erasmus 10,01 03 April 1992 Pretoria
[Adapted from https://worldathletics.org/record]

Use the information above to answ er the questions that follow .

BMI RANGE (in kg/m2)	WEIGHT STATUS
Less than 18,5	Underweight
From 18,5 – 24,9	Normal weight
From 25 - 30	Overweight
More than 30	Obese

ANNEXURE

QUESTION 2.1

LAYOUT PLAN OF THUTO SECONDARY SCHOOL

Mimosa avenue FLOWER STREET
LILLY STREET Entrance for Students Dro p Off/Pick Up
[Adapted from www.sausd.us]

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2024

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including an answer sheet.

TABLE 2: EASTERN CAPE PROVINCIAL BUDGET 2024/2025

SECTORS	2024/2025 (in Rand)	2025/2026 (in Rand)
Education	42 441 422	44 104 248
Health	30 106 843	31 029 965
Social Development	2 972 172	3 100 547
Sports, Recreation, Arts and Culture	1 043 363	1 074 493
Community Safety	144 840	150 179
Total	76 708 640	79 459 432

[Adapted from People's guide to the budget]

2.1.1 Explain the meaning of the term budget.

ANNEXURE A for QUESTION 4.1

Population group	Male Number	% of male population	Female Number	% of female population	Total Number	% of total population
African	21 168 700	80,3	22 165 000	B	43 333 700	80,2
Coloured	2 305 800	8,7	2 465 700	8,9	4 771 500	8,8
Indian/ Asian	677 000	2,6	664 900	2,4	1 341 900	2,5
White	2 214 400	8,4	2 340 400	8,5	4 554 800	8,4
Total	A	100,0	27 636 000	100,0	54 001 900	100,0

ANNEXURE B for QUESTION 4.2

ANNEXURE C for QUESTION 4.3

NATIONAL
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GRADE 1 1

NOVEMBER 2017

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 7 pages including an addendum of 4 pages .

--- Full Combined Text ---

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2025

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages and an addendum with 3 annexures.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ANNEXURES in the ADDENDUM to answer the following questions:

- ANNEXURE A for QUESTION 2.1
- ANNEXURE B for QUESTION 2.2
- ANNEXURE C for QUESTION 4.3

3. Number the answers correctly according to the numbering system used in this question paper.

4. Start EACH question on a NEW page.

5. You may use an approved calculator (non -programmable and non -graphical), unless stated
otherwise.

6. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

7. Round off ALL final answers appropriately according to the given context, unless stated
otherwise.

8. Indicate units of measurement, where applicable.

9. Show ALL calculations clearly.

10.
Write neatly and legibly.

QUESTION 1

1.1 Andrea bakes wedding cakes. The most popular cake she bakes , is a carrot cake with cream cheese frosting. Study the recipe below and answer the questions that follow .

INGREDIENTS
- 350 g self-raising flour
- 325 g dark soft brown sugar
- 1½ teaspoons fine sea salt
- 2 teaspoons cinnamon
- 200 ml olive oil
- 1 teaspoon vanilla extract
- 4 large eggs
- 250 g grated peeled carrots
- 75 g chopped pecan nuts
- 4 tablespoons milk

FOR CREAMY FROSTING
❖ 10 ounces cream cheese
❖ 1¼ cups (140 g) powdered sugar
❖ ¾ cup (80 ml) cold heavy cream
❖ ½ cup salted butter

METHOD
1. Preheat the oven to 356 ■ .
2. Grease and line the bottom and sides of two round cake pans with a double layer of greaseproof paper .
3. Mix the flour, salt, sugar, cinnamon, chopped pecan nuts and grated carrots in a large bowl.
4. Beat together the eggs, olive oil , vanilla extract and milk in a large jug and add to flour mixture. Mix well .
5. Divide the cake batter between the prepared cake pans. Bake for 35 –45 minutes until a skewer comes out clean.
6. Cool the cake layers in the pans for 15 minutes, then carefully turn the cake layers out onto cooling racks.

CREAM CHEESE FROSTING
1. Once the cake layers are completely cooled off , mix the cream cheese and butter in a blender for 1 and a half minutes.
2. Add powdered sugar until frosting is creamy. Whip cold heavy cream into the mixture for 3 minutes until light and fluffy.
3. Cover the cake in frosting – this will take up to 15 minutes.

BAKING CONVERSIONS
1 cup = 250 ml 1 tablespoon = 15 ml 1 teaspoon = 5 ml

[Adapted from https://www.inspiredtaste.net]

1.1.1 Calculate the time (in minutes) to cover the cake with the cream cheese frosting. (2)

1.1.2 Determine the amount of cream cheese needed in the recipe in kilograms.

NOTE: 1 kg = 35,274 ounces (2)

1.1.3 Convert the salt needed in the recipe to millilitres. (2)

1.1.4 Andrea’s oven does not calibrate in degrees Fahrenheit. Determine the temperature, in ■, at which Andrea should set her oven.

You may use the following formula : ■ = ■
■ × (■ - 32■) (2)

1.2 Andrea uses the measuring cup below to measure her ingredients.

[Adapted from www.googleimages.com/measuringcups]

1.2.1 Define the term capacity . (2)

1.2.2 Hence, determine the capacity of the measuring cup. (2)

1.3 Andrea bakes her cakes in round pans, each with a diameter as shown in the diagram below:

[Adapted from www.googleimages.com]

1.3.1 Calculate the circumference of pan Y in squared inches (in.^{2}).

You may use the following formula:

Circumference of a circle = $\pi \times$ diameter , where $\pi = 3{,}142$ (2)

1.3.2 Write the diameter of pan Z to the diameter of pan X as a simplified ratio. (2)

1.4 The map below shows the distance between Andrea's house in Humansdorp and Jeffreys Bay where she needs to deliver a cake for a wedding.

[Adapted from www.googlemaps.com

Use the map above to answer the questions that follow.

1.4.1 Identify the clinic on the map. (2)

1.4.2 In which general direction will Andrea be travelling to Jeffreys Bay? (2)

1.4.3 Convert Andrea's travelling distance to metres. (2)

1.4.4 Determine the time that Andrea will arrive in Jeffreys Bay if she leaves her house at 10:15. (2)

[24]

QUESTION 2

2.1 Pilgrim's Rest is a small museum town in the Ehlanzeni district in Mpumalanga, South Africa.
It has approximately 1 721 residents and occupies an area of 25,40 km^2. The map of Pilgrim's Rest and surrounding areas is shown in ANNEXURE A.

Use ANNEXURE A to answer the following questions.

2.1.1 Identify the tourist attraction found in block D 5. (2)

2.1.2 Determine the probability of finding a railway on this part of the map. (2)

2.1.3 Measure the distance between the Store Museum and the Town Hall in centimetres . (2)

2.1.4 Hence, calculate the actual distance (in kilometres) between the Store Museum and the Town Hall , using the bar scale. (5)

2.1.5 Determine the number of people per square kilometre (km2) residing in Pilgrim's Rest. (2)

2.2 Siphokazi, who resides in Lydenburg , found the map of Mpumalanga and surrounding areas shown in ANNEXURE B.

Use ANNEXURE B to answer the following questions.

2.2.1 Name ONE neighbouring country border ing Mpumalanga. (2)

2.2.2 Identify the number of national roads found on this map. (2)

2.2.3 The actual distance between Lydenburg and Standerton is 276 km and it measures 8,3 cm on the map. Hence, determine the scale of this map. (3)

2.2.4 Siphokazi claims that if she travels at an average speed of 80 km/h and leaves her home at 12:45, she will arrive in Standerton at 16:05, without any stops.
Verify, with the necessary calculations, whether her claim is VALID.

You may use the following formula: Speed = ■■■■■■■■
■■■■ (7)
[27]

QUESTION 3

3.1 Wilmien decided to sell import ed coffee beans in resealable containers to generate funds
in aid of a wheelchair for her physically disabled brother.

The diagram below shows two types of containers that Wilmien can choose from. The containers will be covered in petroleum -based plastic.

RECTANGULAR -SHAPED
CONTAINER CYLINDRICAL -SHAPED CONTAINER

The following formulae may be used:

Volume of a cylinder: π × radius2 × height , where π = 3,142
Volume of a rectangular prism = length × width × height
Total surface area of a rectangular prism: 2 (l × w) + 2 (l × h) + 2 (w × h)

3.1.1 Determine the volume of the cylindrical-shaped container in cubic centimetres (cm3). (3)

3.1.2 Determine the total surface of the rectangular-shaped container that needs to be covered in petroleum-based plastic. Give your answer rounded to the nearest thousand square millimetres (mm2). (6)

3.1.3 The petroleum-based plastic that Wilmien will use to cover the containers is sold at R27,55/m2. Calculate the cost to cover 50 of the rectangular-shaped containers with this plastic.

NOTE: The petroleum-based plastic can only be bought in whole square metres (m2). (4)

3.1.4 Wilmien realised that she cannot fill the containers to maximum capacity. Instead, she fills the containers with coffee beans to a height that is 7% less than the container height.

Show that the height of the coffee beans in the cylindrical container is 11,2 cm. (3)

3.1.5 A sales representative advised Wilmien that the volume of the most economical container should not exceed 1 200 cm3. Wilmien thus claim that the rectangular-shaped container would be the most economical.

Verify, with the necessary calculations, whether Wilmien's claim is VALID. (5)

3.1.6 Provide ONE reason why Wilmien would want to wrap the containers in petroleum-based plastic. (2)
[23]

QUESTION 4

4.1 The layout plan of the shopping mall where Wilmien will be selling her imported coffee beans is shown below. Wilmien will only be selling to the stores that are numbered.

[Adapte d from https://www.za.pinterest.com/pin/400257485606100702/]

Use the layout plan above to answer the following questions.

4.1.1 Determine the probability, as a percentage, that Wilmien will be selling her coffee beans to a store with an uneven number. (3)

4.1.2 Wilmien arrived at the mall to do her deliveries at 08:30 and she concluded her delivery at 10:15. Wilmien claimed that she walked at an average speed of 1,4 miles per hour and thus covered a travelling distance inside the mall of more than 4 km. Verify, with the necessary calculations, whether Wilmien's claim is VALID.

NOTE: 1 mile = 1,60934 km

You may use the following formula: Distance = Speed × Time (7)

4.2 The Pizza Den is a new establishment inside the shopping mall that makes delicious pizzas. A picture of the sliced pizza and its packaging box is shown in the diagram below.

The pizza has a diameter of 42 cm.

[Adapted from www.googleimages.com/pizzaboxdimensions]

4.2.1 Determine the area of the pizza box, rounded to the nearest ten square centimetre s (cm2).

You may use the following formula:

Area of a square = side × side (3)

4.2.2 Determine the value of D. (3)

4.2.3 Calculate the area of one slice of the pizza if it is cut in eight equal slices as illustrated in the picture above.

You may use the following formula:

Area of a circle = π × radius2, where π = 3,142 (4)

4.3 The table used inside The Pizza Den where customers can enjoy their food is shown in ANNEXURE C . The tables are covered with PVC tablecloths with an overhang of 35 cm around the table.

Use ANNEXURE C to answer the following questions.

4.3.1 Calculate the radius of the plastic PVC tablecloth that is used to cover the tables. (3)

4.3.2 Calculate the height of the support pole (in cm) if the total height of the table is 78 cm. (3)
[26]

TOTAL : 100

MARKS: 75

TIME: 1½ hours

This question paper consists of 10 pages.

PROVINCIAL ASSESSMENT

GRADE 11

MATHEMATICAL LITERACY P 2

JUNE 2024

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Start EACH question on a NEW page.

3. Number the answers correctly according to the numbering system used in this question paper.

4. You may use an approved calculator (non-programmable and non-graphical), unless stated otherwise.

5. Show ALL calculations clearly.

6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. Indicate units of measurement, where applicable.

8. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

9. Write neatly and legibly.

QUESTION 1

1.1

Use the information above to answer the questions that follow.

1.1.1 Write down the maximum number of people that this recipe will cater for. (2)

1.1.2 State the recommended time it will take Neo to prepare and cook this dish. (2)

1.1.3 A 1
4 cup of oil is equivalent to 60 ml. Determine the number of millilitres in one cup of oil . (2)

1.1.4 Neo will start to serve this dish at 6:37 pm.

Write down this time using the 24 -hour clock format . (2)
Neo is trying a recipe for making Greek stuffed eggplant. The recipe below shows the ingredients needed for making Greek stuffed eggplant.
PICTURE OF GREEK EGGPLANT INGREDIENTS

Serves
Preparation time: 10 min
Cooking time : 40 min

1.2

1.2.1 Explain the meaning of the scale of the seating plan. (2)

1.2.2 Name ONE other type of scale that you have studied. (2)

1.3 Define the concept outcome as used in probability. (2)
[14]

The scale used on the seating plan of an examination room is 1:50.

QUESTION 2

2.1

Use the map above to answer the questions that follow.

2.1.1 Determine the total number of provinces other than North West shown on this map. (2)

2.1.2 The N4 measures 718 km, starts on the Botswana border and ends in Komatipoort on
the Mozambique border.

Identify any TWO towns shown on the map where the N4 passes through. (2)

2.1.3 The area of the North West province is 104 882 km2. The area of the whole of South
Africa is 1 220 813 km2.

(a) Determine the percentage that the North West province make out of the
whole of South Africa.
(3)

(b) Give a reason why a traveller from Botswana needs a passport to
enter the North West province.
(2)

Below is a map of the North West province showing town/cities and roads.
MAP OF NORTH WEST SHOWING NATIONAL ROADS LEADING TO
TOWNS /CITIES

QUESTION 3

3.1

2.2

Use the map and information above to answer the questions that follow.

2.2.1 State the view represented in the layout picture above . (2)

2.2.2 The distance from Taung to the Pilanesburg Game Reserve is 460 km. The Land Rover covers this distance in 4,5 hours.

Calculate the average speed of this vehicle.

You may use the formula: Distance = speed × time (3)

2.2.3 Calculate the amount of petrol needed for the return trip. (4)

2.2.4 Tumi indicated to his friends that he was 25% sure that he will join them on this trip.

The probability scale below shows the likelihood for various probabilities .

Write down the likelihood that best describes Tumi’s chances of going on this trip. (2)

[20] Mpho, Tshepo and two friends plan to drive to the Pilanesburg Game Reserve near Rustenburg. They will travel from Taung using a Land Rover. The vehicle has a fuel consumption of 6,42 liters/100 km. Its seating layout is shown below.

THE SEATING LAYOUT OF A LAND ROVER

Kea and Tumi are planning their 30th year anniversary this year. Kea wants a four-layered cylindrical vanilla and chocolate cake.

The diagram of the cake is shown below .

KEA’S FOUR LAYERED CYLINDRICAL VANILLA AND CHOCOLATE CAKE

Use the information above to answer the questions that follow.

3.1.1 Determine, in millimetres the total height of the cake. (3)

3.1.2 The cake was baked using 3 and a half pounds of butter.

Determine the mass of butter, in kilograms.

NOTE: 1 pound = 453,592 grams (3)

3.1.3 The bottom layer of the cake has a radius of 14 cm.

(a) Determine, in cm, the diameter of the bottom layer. (2)

(b) Verify, showing ALL the calculations, that the volume of. the bottom layer of this cake is 9 237,48 cm3.

You may use the formula :

Volume of a cylinder = 3,142 × (radius)2 × height. (3)

3.1.4 The cake must be stored in the fridge at 5 oC.
Convert this temperature to oF.

You may use the formula : oF – 32o = (■
■ × oC) (3)

3.2

Mrs Nku has two children, Tom and Neo. They attend two different schools.
The information below describes Mrs Nku's routine on a particular morning.
■ She drives the children to their respective schools .
■ First she drops off Tom at point A.
■ Then she takes Neo to her school at point B.
■ Thereafter she returns home.

Mrs Nku's journey is illustrated in the graph below.

Use the graph and information above to answer the questions that follow.

3.2.1 It took Mrs Nku 15 minutes to drive the 10 km from home to Tom’s school.

Express 15 minutes as a fraction of an hour in decimal form. (2)

3.2.2 Indicate whether the following statements are TRUE or FALSE. If FALSE, correct the statement .

(a) It took Mrs Nku , 25 minutes to drive from Neo’s school to home. (2)

(b) The distance from Tom’s school to Neo’s school is 15 km. (2)

3.2.3 Mrs Nku left her home at 6:47 in the morning to drive her kids to school. On her way back home, she stopped for 20 minutes at the shop to buy bread.

Verify, showing ALL calculations , whether she will arrive at her house before 8:00 a.m. (4)

[24]

QUESTION 4

4.1

Use the information above to answer the questions that follow.

4.1.1 Calculate the circumference of the window.
You may use the formula: Circumference of the circle = 3,142 × d . (2)

4.1.2 Calculate , in cm2, the area of the wall that needs to be painted.

You may use the formulae:

Area of a circle = 3,142 × (■
■)■ where d is the diameter

Area of a square = s2, where s is the length of each side (6)

4.1.3 Convert the answer in QUESTION 4.1.2 to m2. (2)
Sol installed a circular window in the centre of a square wall, as shown in the diagram below. He intends to paint the wall with 3 coats of paint.
The diameter of the circular window is 144 cm .
The length of each side of the square wall is 230 cm.

4.2

Use the diagram above to answer the questions that follow.

4.2.1 Determine the number of offices in this administration building. (2)

4.2.2 The length of the printing room on the scale diagram is 2,2 cm.

Use the given scale above to calculate in metres, the actual length of
the printing room. (3)

4.2.3 Give a reason why the administration building has a waiting area. (2)
(17)

TOTAL: 75

The diagram below shows a scaled diagram of Kitso Secondary School's
administration building.
KITSO SECONDARY SCHOOL'S ADMINISTRATION BUILDING

1:300

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages.

PROVINCIAL ASSESSMENT
GRADE 11

MATHEMATICAL LITERACY P1
NOVEMBER 2024

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Number the answers correctly according to the numbering system used in this question paper.

3. Start EACH question on a NEW page.

4. You may use an approved calculator (non-programmable and non-graphical), unless stated otherwise.

5. Show ALL calculations clearly.

6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. Indicate units of measurement, where applicable.

8. Diagrams are NOT necessarily drawn to scale, unless stated otherwise.

9. Write neatly and legibly.

QUESTION 1

1.1

Use the cash register slip above to answer the questions that follow.

1.1.1 How many items are purchased in total on the cash register slip ? (2)

1.1.2 During which month is this purchase made according to the cash register slip ? (2)

1.1.3 Define the term VAT . (2)

1.1.4 Show how the R24,34 VAT amount is calculated . (2)

1.1.5 Is this purchase paid with card or cash ? (2)

1.1.6 Write the total amount due in words . (2)

JANE’S SUPERMARKET
Welcome to our store !
A9 Daven Avenue , East London
Telephone number : 043 711 1234
BTW Registration No.: *** 951753
Coca Cola 2 ■
Simba chips 125 g – salt & vinegar
Yoghurt 6 pack – strawberry
All Gold tomato sauce
OMO washing powder 2 kg
TOTAL
CASH
CHANGE R24,95
R15,90
R21,90
R35,90
R87,95
R186,60
R200,00
R13,40
VAT 0% R0,00
VAT 15% R24,34
25/09/2024 16:10 Cashier : Belinda

1.2

Use the graph above to answer the questions that follow.

1.2.1 Name the type of graph used to represent the information above. (2)

1.2.2 What percent romance movies is watched? (2)

1.2.3 Which genre is the most watched according to the graph above ? (2)

1.2.4 Determine the probability that a movie watched is a true -life story. (2)
[20]

Comedy
28%
Action
20%RomanceDrama
14%Horror
12%Sci-Fi 8%DIFFERENT GENRES OF MOVIES
WATCHED

QUESTION 2

2.1 The Smith family would like to go on vacation in 2025. Mr. Smith examines their finances per month in 2024 and sets up the following monthly budget to determine how much they can save for their vacation the following year .

TABLE 1: MONTHLY BUDGET FOR THE SMITH FAMILY

INCOME		EXPENSES	
Mr. Smith salary (Manager)	R25 000	Home loan	R8 400
Mrs. Smith salary (Teacher)	R17 000	Car instalments	R8 200
Rent income	R5 000	Insurance	R2 100
		Groceries (R1 150 per week)	
		Fuel	R4 500
		Entertainment	R2 000
		School fees and activities	R4 850
		Medical expenses	R1 850
TOTAL INCOME	R47 000	TOTAL EXPENSES	.

Use the information above to answer the questions that follow .

2.1.1 Calculate how many litres of fuel the Smith family uses per month if the price of fuel is R23,25 per litre. (2)

2.1.2 Calculate the total groceries for the month. (2)

2.1.3 Calculate how much money the Smith family has left at the end of the month to save for their vacation. (4)

2.1.4 If the rent income they received per month in 2023 was R 4 465 , calculate , to the nearest whole number, the percentage increase from 2023 to 2024 .

You may use the following formula:

% increase = ■■■■ ■■■■■■ ■■■■ − ■■■■ ■■■■■■ ■■■■
■■■■ ■■■■■■ ■■■■ × 100% (3)

2.1.5 Name ONE other type of income that a household can earn. (2)

2.2
The Smith family is considering the following travel package to the Kruger National Park and it includes the following:

- ■ Scheduled return flights from Johannesburg to the Kruger
- ■ All road transfers between airport and activities
- ■ 2 nights ' accommodation at Sudwala Lodge
- ■ 4 nights ' accommodation at Sabie River Bush Lodge
- ■ All breakfast and dinners included
- ■ Elephant interaction where you and the little ones can feed, greet and walk hand -in trunk with the gentle giants
- ■ Early morning and late afternoon off-road game drives that take place in an open 4×4 safari vehicle with a ranger and tracker team
- ■ Child friendly facilities and kids activities
- ■ Park and conservation fees

The cost for this package is from R 24 999 per person sharing and is only available during low season.

Use the information above to answer the questions that follow .

2.2.1 Write the cost of the package in words . (2)

2.2.2 Write in simplified form, the ratio of nights' accommodation at Sudwala Lodge to the nights' accommodation at Sabie River Bush Lodge . (2)

2.2.3 Name ONE other expense that may be incurred during their holiday that is not included in the package . (2)

2.2.4 Mr. Smith deposits R5 500 every month for the next 12 months in a savings account for their holiday. Every month he earns 7% interest on the amount he deposits. Calculate how much money he will have in total in his savings account after a year . (5)

2.2.5 Mrs. Smith invests R30 000 in a savings account and earns 9,2% interest per year, compounded bi -annually.

She claims that if she adds her money from her savings account together with her husband's money from his savings account after one year, they will have enough for 4 people to go on vacation to the Kruger National Park according to the travel package above.

Verify, by showing ALL calculations, whether her claim is CORRECT . (7)
[31]

QUESTION 3

3.1 Athletics consists of different events. One of these events is the 100 metre . The table below shows the ten fastest times ever for men of South Africa in the 100 metre event .

TABLE 2: 10 FASTEST TIMES OF THE MEN'S 100 METER EVENT IN SOUTH AFRICA

	NAME AND SURNAME	Time (sec)	Date of birth	Place of birth
1	Akani Simbine	09,82	21 September 1993	Kempton Park
2	Benjamin Richardson	09,86	19 December 2003	Pretoria
3	Shaun Maswanganyi	09,87	01 February 2001	Soweto
4	Henricho Bruintjies	09,89	16 July 1993	Paarl
5	Retshidisitswe Mlenga	09,93	27 February 2000	Bloemfontein
6	Wayde van Niekerk	09,94	15 July 1992	Kraaifontein
6	Gift Leotlela	09,94	12 May 1998	Phuthaditjhaba
8	Thando Roto	A	26 September 1995	King Williams Town
9	Simon Magakwe	09,98	14 May 1986	Itsoseng
10	Emile Erasmus	10,01	03 April 1992	Pretoria

[Adapted from https://worldathletics.org/record]

Use the information above to answer the questions that follow .

3.1.1 Which athlete's times is the 4th fastest time ever from South Africa ? (2)

3.1.2 Arrange the date of birth of the 10 athletes from the oldest to the youngest . (2)

3.1.3 Determine the modal place of birth of the athletes . (2)

3.1.4 Calculate A, the time of Thando Roto , if the average time is 09,919 seconds for the top 10 men in the 100 metre event . (4)

3.1.5 Calculate the range of the 100 metre times . (3)

3.2 The graph below shows the nine provinces ' medals won during the national high school athletics meeting held in 2024 .

Use the information above to answer the questions that follow .

3.2.1 How many medals did Gauteng won in total ? (2)

3.2.2 Determine the difference between North West and Free State 's silver medals . (3)

3.2.3 What percentage of Western Cape 's total number of medals won was bronze during the 2024 national meeting ? (3)

3.2.4 Determine the probability of how many of the 9 provinces received more than 20 gold medals. Give your finale answer as a decimal number , rounded to TWO decimal places. (3)

3.2.5 In 2023, during the national high school athletics meeting , Limpopo province only won 8 medals in total . It is claimed that their percentage growth from their total medals has increases by more than 50% .

Verify, by showing ALL calculations , if the claim is VALID . (4)
[28]

29
201610 8 6 5 4 323
1818
977 86 517
25
18
12
6 7 466MEDALS WON DURING THE 2024
NATIONAL HIGH SCHOOL ATHLETIC
MEETING
Gold Silver Bronze

QUESTION 4

4.1 Mrs. Molefe investigates his bank charges for a certain month. She looks at what transactions she has made during the month to determine how much each transaction 's bank charges should be . The table below shows certain transaction costs of his bank .

TABLE 3: TRANSACTION COSTS

TRANSACTIONS	TRANSACTION COSTS
Cash withdrawal from own bank's ATM	1,15% of the amount withdrawn
Cash withdrawal from another bank's ATM	R9,50 + own bank's transaction costs
Deposit at own bank's ATM	2,25% of amount deposited
Debit order /Stop order	R6,00 per transaction

.

Use the information above to answer the questions that follow.

4.1.1 Mrs. Molefe 's opening balance for the month is R42 952,19. On the first of the month, the following debit orders were deducted from her account :

- Insurance – R2 100,00
- Home loan instalment – R8 800,00

Calculate her balance at the end of the first day of the month . (3)

4.1.2 On the 2nd of the month, an amount of R1 150 was withdrawn from her own bank's ATM. Calculate Mrs. Molefe 's balance at the end of the 2nd of the month . (4)

4.2 Mrs. Molefe wants to buy a new motor car. She uses the loan factor table below to determine the interest rate against the loan period .

TAB LE 4: LOAN FACTOR TAB LE

LOAN PERIOD	INTEREST RATE						
	10%	10,25 %	10,50 %	10,75 %	11%	11,50 %	12,00 %
3 years	32,27	32,38	32,50	32,62	32,74	32,98	33,21
4 years	25,36	25,48	25,60	25,72	25,84	26,09	26,33
5 years	21,25	21,37	21,49	21,62	21,74	21,99	22,24
6 years	18,53	18,65	18,78	18,91	19,03	19,29	19,55

Selling price	R198 000
Deposit	R29 700
Interest rate	10,50%
Terms	48 m onths

.

Use the information above to answer the questions that follow.

4.2.1 Determine the probability that the loan factor will be more than 19,00 % for the loan period of 6 years . Give your finale answer as a percentage, rounded to the nearest whole number . (3)

4.2.2 Calculate the loan amount that Mrs. Molefe has to borrow for the motorcar once the d eposit is taking into account . (2)

4.2.3 Use the loan amount (QUESTION 4.2.2) and the table of loan factors above to calculate h er total monthly payment , plus R65,00 administration fee .

You may use the following formul a:

Monthly payment = ■■■■ ■■■■■■
■ ■■■ × loan factor (3)

4.2.4 Mrs. Molefe makes a statement that she will pay more than R40 000 of interest if she buys the motorcar with the 48-month option .

Verif y, by showing ALL calculations , whether h er statement is CORRECT . (6)
[21]

GRAND TOTAL: 100

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MARKS: 100

TIME: 2 hours

This question paper consists of 1 2 pages and an annexure .

PROVINCIAL ASSESSMENT
GRADE 11

MATHEMATICAL LITERACY P 2

NOVEMBER 2024

INST RUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the attached ANNEXURE for QUESTION 2.1.

3. Start EACH question on a NEW page.

4. Number the answers correctly according to the numbering system used in this question paper.

5. You may use an approved calculator (non -programmable and non -graphical), unless stated otherwise.

6. Show ALL calculations clearly.

7. Round off ALL final answers appropriately according to the given context, unless stated
otherwise.

8. Indicate units of measurement, where applicable.

9. Diagrams are NOT necessarily drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

QUESTION 1

1.1

Use the measuring tools abovu to answer the questions that follow.

1.1.1 (a) Name the instrument labelled A. (2)

(b) Define the concept temperature . (2)

(c) Write , in oC, the temperature at the black line on the instrument labelled A. (2)

1.1.2 Convert the volume of the substance in the container B to litres. (2)

1.1.3 Determine, in mm, the distance between the two arrows on the ruler in diagram C. (3)

The following measuring instruments are given below:

A.

B.

C.

Use the map above to answer the questions that follow.

1.2.1 Identify the type of map shown above. (2)

1.2.2 Determine , in metres, the actual distance between Johannesburg and Villiers. (2)

1.2.3 Name the town that is 166 km from Durban. (2)

1.2.4 Write down the total number of towns on the N3 that Lelo must pass on her way to Durban. (2)

[19]

Lelo will be travelling from Johannesburg to Durban for the holidays. She will use the
map below to plan her journey.

MAP OF THE ROUTE FROM JOHANNESBURG TO DURBAN

QUESTION 2

2.1

Use the ANNEXURE and the information above to answer the questions that follow.

2.1.1 Identify the room that is between the staff room and the tuck shop. (2)

2.1.2 Use the compass direction to determine on which sides of the school hall, the doors are situated. (2)

2.1.3 The two music classrooms are stand -alone (not attached) classrooms.

(a) Write down the numbers of these classrooms. (2)

(b) Describe the location of the music classrooms in relation to the streets shown on this layout plan. (3)

2.1.4 A learner followed these directions to locate his classroom:

- From the main office he walked in a southerly direction.
- He turned right after passing the quad and continued straight.
- The second last classroom on his left -hand side was his classroom.

Write down the learner’s classroom number. (2)

2.1.5 Give ONE reason why Mimosa Avenue was designed as a one -way road. (2)

2.1.6 Calculate, as a percentage , rounded to THREE decimal places, the probability of finding a classroom with its door on the western side. (4)

2.2 Learners from Thuto Secondary School uses different modes of transport to get to school.

2.2.1 Solly, one of the learners, travels 12 km in 60 minutes to get to school.

Calculate , in metres per minute (m/min), Solly’s average speed .

You may use the formula: Distance = Average speed × Time (4)

2.2.2 Use Solly’s average speed from QUESTION 2.2.1 to determine whether he was walking or using any other mode of transport. Give a reason for your answer. (3)

[24] The ANNEXURE shows the layout plan of Thuto Secondary School. Besides the classrooms and other buildings, the school has two science laboratories (Sci 1 and Sci 2) and a quad which is the open -air space for assemblies.

QUESTION 3

3.1

Use the information above to answer the questions that follow.

3.1.1 Calculate, in m2, the area of 1 bicycle.

You may use the formula: Area of a bicycle = length × width (3)

3.1.2 The total floor area required for the 127 bicycles is 166,37 m2.

Verify, showing ALL calculations, how the floor area was determined . (4)

Qhubeka Charity Organisation donated 127 bicycles to all learners of Thuto Secondary School who took longer than 1 hour each day to walk to school. A second organisation provided bicycle stands for the 127 bicycles.

The rectangular floor area that are required for the parking of the bicycles is determined by considering the following:

- Each bicycle is 180 cm long and 45 cm wide.
- An additional space of 0,5 m2 per bicycle is required for free movement around the stands.

Bicycle
stand s

3.2 The SGB of Thuto Secondary School decides to pave the floor area where the learners will park their bicycles. They will use paving bricks like the one shown below:

INFORMATION
ABOUT THE BRICK S THE BRICK PALLET OF BRICKS
Number of bricks per
1 m2 : 48

Paving bricks are sold in
pallets of 594 bricks

Weight of 1 brick: 2,230 kg

3.2.1 The SGB will buy extra 5% of bricks for wastage and breakage .

Determine the total number of bricks that the SGB will buy to pave the floor area where the bicycles will be parked. (5)

3.2.2 Determine the number of pallets of bricks that the SGB must buy. (3)

3.2.3 Determine, correct to ONE decimal place, the weight in tonnes of all the bricks needed.

NOTE : 1 ton = 1 000 kg (4)

Please turn over 3.3

Use the graph above to answer the questions that follow.

3.3.1 Identify and explain the type of relationship represented by the graph above. (3)

3.3.2 Derive the formula to show the relationship above. (2)

3.3.3 Use the formula in QUESTION 3.4.2 or otherwise to calculate P, the number of days taken by 8 workers to complete the project. (2)
[26]

The SGB will employ workers from the community to pave the floor area.

The graph below shows the relationship between the number of workers they employ, and the time taken to complete the paving project.

05101520253035
1 2 3 4 5 6 8 10 12 15 20 30Number of days
Number of workersTHE NUMBER OF WORKERS VS THE TIME TAKEN TO PA VE THE AREA
P

QUESTION 4

4.1 Below are the assembly diagrams for a floor lamp.

ASSEMBLY DIAGRAMS FOR A FLOOR LAMP

Use the diagram above to answer the questions that follow.

4.1.1 Refer to DIAGRAM 4.

Give t he direction in which the nut should be turned . (2)

4.1.2 Which diagram is associated with the instruction: ‘Join the stand to the base?’ (2)

4.1.3 The actual height of the floor lamp is 1,55 m.

Determine, in cm, the height of the floor lamp in the diagram if the scale is 1 : 25 (3)

1 2 3 4
5 6 7 8

4.2

Use the information above to answer the questions that follow.

4.2.1 Define the concept volume. (2)

4.2.2 Calculate, in m3, the volume of the box.

You may use the formula: Volume of a box = length × width × height (3)

4.2.3 Peter drives a delivery van as the one shown below:

DIMENSIONS OF THE
DELIVERY VAN THE DELIVERY VAN
length: 1 700 mm

width: 1 490 mm

height: 1 200 mm

Calculate the maximum number of boxes that can be packed into the van as follows:

- the length of the box along the length of the van,
- the width of the box along the width of the van and
- then boxes are stacked to the roof of the van. (6)

4.2.4 Peter picked up the parcels at 14:50 from the warehouse on 30 June and delivered one parcel at 08:15 on 2 July.

Verify , showing ALL calculations, whether this delivery was done within the specified delivery time of 48 hours. (4)

Peter works for a courier company in Rustenburg. The company collects packages , in boxes, from different warehouses across the country and delivers them to clients within
48 hours . Below are the dimensions of a box, that must be delivered.
DIMENSIONS OF A BOX THE BOX
Length: 340 mm
Width : 325 mm
Height: 180 mm

4.3

Use the information above to answer the questions that follow.

4.3.1 Write the acronym BMI in full. (2)

4.3.2 State Tshidi's current weight status. (2)

4.3.3 Calculate Tshidi's current mass.

You may use the formula: BMI = ■■■■ ■■ ■■
(■■■■■■ ■■ ■)■ (3)

4.3.4 Provide an advice to Tshidi on how she can improve her weight status. (2)
[31]

TOTAL: 100

Tshidi is a soccer player, and she is concerned about her weight. Her current BMI is 18,2 kg/m2 and a height of 1,56 m.
The table below shows the weight status versus the BMI range.
TABLE: WEIGHT STATUS ACCORDING TO BMI.

BMI RANGE (in kg/m2)	WEIGHT STATUS
Less than 18,5	Underweight
From 18,5 – 24,9	Normal weight
From 25 - 30	Overweight
More than 30	Obese

ANNEXURE

QUESTION 2.1

LAYOUT PLAN OF THUTO SECONDARY SCHOOL

Mimosa avenue FLOWER STREET
LILLY STREET Entrance for Students Dro p Off/Pick Up
[Adapted from www.sausd.us]

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2024

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including an answer sheet.

Imlit1

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.

2. Answer ALL questions.

3. Use the ANSWER SHEET provided to answer QUESTION 4.1.3 and 4.1.4.

4. Start EACH question on a NEW page.

5. Number the answers correctly according to the numbering system used in this question paper.

6. Leave ONE line between two sub-questions, for example between QUESTION 2.1 and QUESTION 2.2.

7. You may use a non-programmable calculator.

8. You may use appropriate mathematical instruments.

9. Show ALL formulae and substitutions in ALL calculations.

10. Round off ALL final answers TWO decimal places, unless stated otherwise.

11. Write neatly and legibly.

QUESTION 1

1.1 Mrs Link ordered clothing items and received the invoice below.
The invoice for purchasing of clothes.

CANVA ©
INVOICE
BILLED TO:
Ave Link
Invoice No: 2468
093 753 8800
16 February 2024
6 Prince Road

ITEM	UNIT PRICE	QUANTITY	PRICE
White camisole top	$123	1	$123
Cuban collar shirt	$127	4	A
Floral cotton dress	$123	B	$369
Subtotal			$1 000
Tax (0%)			$0
TOTAL			$1 000

Thank you

PAYMENT INFORMATION

Briand Bank
Account number:123 -456-8900 Samira Hadid
Pay by: 5 March 2024 331 Anywhere
St. Any City
ST 32110

Use the above information to answer the questions that follow.

1.1.1 Write down the name of the service provider. (2)

1.1.2 Calculate the value of A. (2)

1.1.3 Define the term invoice. (2)

1.1.4 Determine the value of B. (2)

1.1.5 Write down the due date? (2)

1.1.6 Convert the total amount in dollars to rand, if R1,00 = $0,053 . (2)

Copyright reserved Please turn over1.2 Metro church members are planning to have a church service at a resort for a weekend.
They received quotes from two different resorts, both which accommodate a maximum of 10 people per resort.
Mt View Resort costs R1 200 per night including breakfast.
Forest Resort costs R950 per night.
The above costs are shown in the table below.
TABLE 1: COSTS FOR BOTH RESORTS

Number of people	0	1	C	4	8	10
Cost for Mt View Resort	1 200	1 200	1 200	1 200	1 200	1 200
Cost for Forest Resort	950	950	950	950	950	D

Use the above information to answer the questions that follow.
1.2.1 Identify the type of relationship illustrated in the table above. (2)
1.2.2 Calculate how much the church members will pay if only five members attend the service at Mt View Resort. (2)
1.2.3 Determine the following missing values in the table:
(a) C (2)
(b) D (2)
1.3 A Grade 11 Mathematical Literacy class wrote a March cycle test. The results of the cycle test are shown in the graph below.
1.3.1 Determine the number of learners that wrote the cycle test. (2)
1.3.2 Calculate the percentage of learner s who failed the cycle test. (2)
1.3.3 Write down the modal class percentage levels. (2)
[26]29
23
34
05101520253035
0 ■ 29 30 ■ 49 50 ■ 69 70 ■ 100NUMBER OF LEARNERS
PERCENTAGE LEVELS2024 GRADE 11
MARCH CYCLE TEST RESULTS

QUESTION 2

2.1 Study the Eastern Cape P rovincial Budget 2024/2025 framework and its focus areas on
sectors. The summary of payment and estimates per sector is shown in the table below.
Some information has been omitted.

TABLE 2: EASTERN CAPE PROVINCIAL BUDGET 2024/2025

SECTORS	2024/2025 (in Rand)	2025/2026 (in Rand)
Education	42 441 422	44 104 248
Health	30 106 843	31 029 965
Social Development	2 972 172	3 100 547
Sports, Recreation, Arts and Culture	1 043 363	1 074 493
Community Safety	144 840	150 179
Total	76 708 640	79 459 432

[Adapted from People's guide to the budget]

2.1.1 Explain the meaning of the term budget. (2)

2.1.2 Write down the amount allocated for health in 2025/2026 in words. (2)

2.1.3 Calculate the percentage increase of the Education sector for the 2024/2025 and 2025/2026 financial year. Give your answer to the nearest percentage.

You may use the following formula:

■■■■■■■■■■ ■■■■■■■■ = ■■■ ■■■■■ ■ ■■■ ■■■■■
■■■ ■■■■■×■■■ % (4)

2.2 Mr Paul is a businessman. He wants to attend a meeting and searched a meter taxi online.
He found the prices of two companies: Daniel's company and Jerry's company.

Daniel's company charges the following for a single trip:

x A minimum call-out fee of R50 per trip and the first three kilometres are free
x Thereafter, R12,00 for each additional kilometre or part thereof

Jerry's company charges a flat rate of R14,50 per kilometre.

Use the above information to answer the questions that follow.

2.2.1 Write down the formula for calculating the cost for Daniel's company in the form of:

Cost = (2)

2.2.2 The distance for a single trip to the meeting is 80 km. Mr Paul makes the statement that Jerry's taxi company will be the cheapest. Verify, showing ALL calculations, that his statement is correct.
(6)

2.2.3 Mr Paul had another trip. He paid Daniel R1 214 for a single trip. Determine the distance travelled during this trip. (4)

2.3 Mr Mnotho and his wife both secured new jobs overseas. They applied at a Boys High School for their two sons. The first son will be doing Grade 10 and the second son will be doing Grade 8. They will be staying at the hostel. Mr Mnotho received the following information regarding the school fees.

TABLE 3: SCHOOL FEE STRUCTURE FOR 2024

SCHOOL FEES	R48 800,00
HOSTEL FEES	R66 800,00
TOTAL	R115 600,00

School fees must be paid before end of January.
Hostel fees must be paid at the beginning of each year.
Discount for siblings
is offered as follows:
x Second child gets 10% discount on school fees and 5% discount on hostel fees
x Third child gets 15% discount on school fees and 10% discount on hostel fees

Calculate the total discount amount that Mr M notho and his wife will get from the school. (6)

[26]

QUESTION 3

Study TABLE 4 below showing crypto ownership of the top 13 countries in 2023 and answer the questions that follow.

TABLE 4: TOP 13 COUNTRIES WITH CRYPTO CURRENCY

No.	COUNTRY	VALUES	PERCENTAGES (%)
1.	Indonesia	12 615 365	4,55
2.	Colombia	2 505 605	4,81
3.	Morocco	1 854 162	4,90
4.	Kenya	2 713 117	4,92
5.	Turkey	4 684 727	5,54
6.	United Kingdom	3 740 280	5,52
7.	Argentina	2 544 102	5,56
8.	Nigeria	12 862 740	5,75
9.	Russia	8 485 749	5,87
10.	Pakistan	15 400 547	6,40
11.	Brazil	15 400 547	6,40
12.	India	103 317 638	7,23
13.	Thailand	6 692 796	9,32
	TOTAL		

[Source: za.pinterest.com]

3.1 Write down the country with the highest value of crypto currency. (2)

3.2 Arrange the values owned by the countries in descending order. (2)

3.3 Calculate the difference between the maximum value and the minimum value. (2)

3.4 Determine the number of countries with a percentage between 5% and 6%. (2)

3.5 Determine the median country of the top 13 countries. (2)

3.6 Name the type of graph that would best display the above information. (2)

3.7 Calculate the probability, as a decimal, of randomly selecting a percentage between 3% and 6%.
(3)

3.8 Write down the number of countries from the list of top 13 countries with crypto currency that are found in Africa. (2)

[17]

QUESTION 4

4.1 The learners of the Student Christian Association (SCA) of South Africa are planning to celebrate the 128th year of existence of the organisation in October 2024. The Student Christian Association (SCA) was founded in 1869 through missionary work in Stellenbosch. Celebrations will take place in the Eastern Cape . The Eastern Cape region of SCA learners, as the hosting province, are planning to have a fundraising event. They intend to use one of the biggest stadiums in the province as the venue.

Their planning is as follows:

x Stadium	R50,00 per learner
x Decorations	R2 400,00

The table below shows the cost of using the stadium.

TABLE 5: COST OF USING THE STADIUM

Number of learners	0	10	20	40	50	100
Cost in Rand	2 400	2 900	3 400	4 400	4 900	7 400

After calculating all the expenses, they decided to fundraise by selling t-shirts that will be worn during the celebrations, at R150,00 each.

The table below shows the income from selling the t-shirts.

TABLE 6: INCOME FROM SELLING T-SHIRTS

Number of learners	0	10	20	40	50	100
Costs in Rand	0	1 500	3 000	6 000	7 500	15 000

Use TABLE 5 and 6 and the information above to answer the questions that follow.

4.1.1 The cost of using the stadium is expected to increase in 2025. Write down the term used to explain the yearly general increase of goods or services. (2)

4.1.2 Show how the total cost of using the stadium when 100 learners booked to attend the session, is calculated. (3)

4.1.3 Use TABLE 6 to draw a line graph of the income from selling t-shirts, on the axis provided on the ANSWER SHEET. (3)

4.1.4 Indicate on the graph with a letter B, a point that will determine the number of t-shirts the SCA learners must sell to break even. (2)

4.1.5 After the fundraiser the learners made a profit of R7 500 and they deposited it in an investment account, earning interest of 10,5% p.a. compounded annually. The student body claims that after 3 years they will have more than R10 000 in the account. Verify, showing ALL calculations, whether the statement is valid. (5)

4.2 The 2023 Grade 11 Mathematical Literacy class of Zulu High School wrote an examination marked out of 100 marks. The results are shown in the table below.

43	32	86	39	30	26	62	39	40	41	50
21	18	38	26	33	69	17	22	51	21	37
24	12	24	41	54	41	E	14	15	55	49
64	60	49	15	40	44	53	31	13	23	26

Use the information above to answer the questions that follow.

4.2.1 Is the above data continuous or discreet? (2)

4.2.2 The mean of the above data is equal to 37. Calculate the value of E. (5)

4.2.3 The Mathematical Literacy teacher, Mrs Lee, states that the results shown in the table above are more clustered together. Verify, with the necessary calculations, whether her statement is correct.
(4)

4.2.4 Determine the number of learners who achieved 30% or more. (2)

4.2.5 Write down the probability of selecting learners who achieved more than 80% as a decimal. (Correct your answer to 3 decimals places.) (3)
[31]

TOTAL: 100

ANSWER SHEET
QUESTION 4.1.3 and 4.1.4
NAME OF LEARNER: ……………………………………….
010002000300040005000600070008000900010000110001200013000140001500016000
0 1 02 03 04 05 06 07 08 09 0 1 0 0110Cost in Rands
Number of Learners

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2024

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages, and an addendum with 2 annexures.

imlit2

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions.

2. Use the ANNEXURES in the ADDENDUM to answer the following questions:

x ANNEXURE A for QUESTION 2.1
x ANNEXURE B for QUESTION 4.2

3. Answer ALL the questions.

4. Number the answers correctly according to the numbering system used in this question paper.

5. Start EACH question on a NEW page.

6. You may use an approved calculator (non-programmable and non-graphical), unless stated otherwise.

7. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

8. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

9. Indicate units of measurement, where applicable.

10. Show ALL calculations clearly.

11. Write neatly and legibly.

QUESTION 1

1.1 Zanele and Reneiloe are driving to Pretoria one morning. Zanele departs from Randburg and picks up Reneiloe who stays in Midrand. The car is driven at an average speed of 100 km/h from Randburg to Midrand and then to Pretoria. The clock below shows the time at which Zanele left Randburg.

[Source: www.googleimages.com]

1.1.1 Identify the type of clock displayed above. (2)

1.1.2 Write down the time Zanele left home in words. (2)

1.1.3 Zanele arrived at Reneiloe's place at 10:27. Determine how long it took her to get to Midrand. (2)

1.2 Sisanda drew the diagram below, showing the room she uses to study.

1.2.1 Define the term perimeter . (2)

1.2.2 Hence, calculate the perimeter of the room in centimetres. (2)

1.3 Timothy wants to lay ceramic tiles in his rectangular bedroom. Study the details of the size of the tiles that Timothy will use below and answer the questions that follow.

[Adapted from www.googleimages.com]

1.3.1 Convert 43 cm to metres (m). (2)

1.3.2 Hence, calculate the area of the tile in m2.
You may use the following formula: Area of a square = side × side (2)

1.4 Study the list of items below that are needed to assemble an office chair and answer the questions that follow.

1.4.1 Identify the number of casters (wheels) needed to assemble the office chair. (2)

1.4.2 Name the tool that will be used to tighten the bolts and washers. (2)

1.4.3 List the different types of bolts and washers needed for the backrest. (2)

[20]

QUESTION 2

2.1 Knysna is a town in the Western Cape province of South Africa. The map of Knysna and surrounding areas in the Western Cape is shown in ANNEXURE A.

Use ANNEXURE A to answer the questions below.

2.1.1 Identify the national road on the map. (2)

2.1.2 In which general direction will you go if you travel from Belvidere Village to Leisure Island? (2)

2.1.3 The actual distance between Paradise and Hornlee is 7 km. Use the scale to calculate the map distance (to the nearest cm) between the two towns. (4)

2.2 Busisiwe lives in Knysna Heights and plans a trip to Pezula Estate where she will spend the weekend with her friends. The distance between Knysna Heights and Pezula Estate is 11,2 km travelled via the N2.

Busisiwe travels at an average speed of 100 km/h and she drives a Polo Vivo 1.4 with a fuel consumption of 5,7 litres per 100 km.

Use the map of Knysna and the above information to answe r the following questions.

2.2.1 Calculate the approximate time, in minutes , that Busisiwe will take to travel from Knysna Heights to Pezula Estate.

You may u se the following formula: Speed = ■■■■■■■■
■■■■ (4)

2.2.2 Busisiwe and her friends travelled an approximate distance, in and around Pezula Estate, of 156 km with her car. Determine the total distance Busisiwe travelled during the weekend including he r trip to Pezula Estate and back. (3)

2.2.3 Busisiwe claims that she will spend less than R250,00 on petrol for her return trip including the weekend driving to and from Pezula Estate.

Verify , with the necessary calculations, whether her statement is valid or not.

NOTE: Cost of petrol = R24,45 per litre (5)
[20]

QUESTION 3

3.1 Hein bought a fish tank at Premium Aquatics. The fish tank, with dimensions, is shown in the picture below.

[Source: https://aquapap.com/fish-tanks]

Use the above information to answer the questions that follow.

3.1.1 Define the term volume in the context above. (2)

3.1.2 Calculate the height of the fish tank if the volume is 38,8 ft3.

You may use the following formula:

Volume of fish tank = π × radius2× height , where π = 3,142 (3)

3.1.3 The required temperature for the fish tank is 72 ■. Hein claims that he needs to set the temperature to 22,2 ■ to be equivalent to the required temperature.

Verify, with the necessary calculati ons, whether Hein's claim is valid or not.

You may use the following formula: ■= (■-3 2o) × ■
■ (3)

3.1.4 The fish tank is 75% full of water. Af ter adding stones to the bottom of the fish tank, the fish tank is 87% full of water. Calculate the volume of the stones in cubic feet (ft
3). Round your final
answer to ONE decimal place. (5)

3.1.5 Hein has 7 fish in his fish tank: 3 yellow, 1 blue, 2 red, and 1 silver. Determine the probability (as a percentag e) of selecting a silver fish from the fish tank. (3)

3.2 Hein wants to build a cabinet to display hi s fish tank and to prevent it from falling and breaking. His bricklayer informed him that he would need about 2 500 bricks for the cabinet he has in mind. Below is an image of the brick they will use.

3.2.1 Calculate the total surface area (in mm2) of one of the bricks needed for the wall display cabinet.

You may use the following formula:
Total Surface Area = [2 (l × w) + 2 (l × h) + 2 (w × h)] (4)

3.2.2 Determine the number of complete pallets transported by truck if one pallet contains 500 bricks. (2)

3.2.3 The weight of the pallets transported is 9 187,5 kg. Convert the weight of one pallet to ton.

NOTE: 1 000 kg = 1 ton (3)

3.2.4 Hein claims that the volume of all the bricks is 4,7 m3. Verify, with the necessary calculations, whether Hein's claim is valid or not.

You may use the following formula:
Volume of one brick = Length × Width × Height (5)

3.2.5 One of the bricks that Hein is going to use costs R2,60 and the cost of delivery is R650.

Calculate the cost Hein will have to pay for the bricks that will be needed, including delivery. Round your final answer to the nearest hundred rand. (4)
[34]

QUESTION 4

4.1 Scranton is a hamlet in the town of Pennsylvania in New York, United States. Study the road map of Scranton in ANNEXURE B of the addendum and answer the questions below.

4.1.1 List TWO types of scales used on maps. (2)

4.1.2 Using the given scale, calculate the actual distance in miles between Clarks Summit and Archbald, if the distance on the map between these two places is 10,5 cm. (5)

4.1.3 A tourist must attend a conference in Blakely at 15:00. The travelling distance from his house to Blakely is 121,4 miles. He claims that if he leaves his house at 13:30 and stops at the petrol station for 25 minutes, he will be on time for his conference . The tourist travels at an average speed of 85 miles/hour.
Verify , with the necessary calculations, whether his claim is valid or not.
You may use the following formula:
Time =■■■■■■■■■
■■■■■(6)

4.1.4 Determine the probability of randomly selecting a road on the map that is a state road. (2)

4.2 Lwandile bought an Ugandan flag mounted on a rectangular wooden frame as shown in thediagrams below.
(Diagram NOT drawn to scale).
PICTURE OF THE UGANDAN
FLAGDIMENSIONS OF THE
RECTANGULAR WOODEN FRAME

4.2.1 Show that the area of the white cloth needed to cover the circle is 70,89 cm2.
You may use the following formula:
Area of a circle = π × radius2, where π = 3,142 (3)

4.2.2 Determine the width of the rectangular wooden frame in centimetres, if the area of the rectangular wooden frame, without the circle, is 2 682 cm2.

You may use the following formula:
Area of a rectangle = Length × Width
NOTE: 1 inch = 2,54 cm (5)

4.2.3 Lwandile claims that the width of ONE of the rectangular bars is 8 cm.
Verify, with the necessary calculations, whether his claim is valid or not.
(3)
[26]

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2023

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including an answer sheet.

Imlit1

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ANSWER SHEET to answer QUESTION 4.1.4.

3. Number the answers correctly according to the numbering system used in this question paper.

4. Diagrams are not necessarily drawn to scale, unless stated otherwise.

5. Round off ALL final answers according to the context used, unless stated otherwise.

6. Indicate units of measurement, where applicable.

7. Start EACH question on a NEW page.

8. Show ALL calculations clearly.

9. Write neatly and legibly.

QUESTION 1

1.1 David runs a small motorcycle service business from home. Given below is the budget drawn up by him for the new month.

INCOME	AMOUNT	EXPENSES	AMOUNT
26 Motorcycle services	R33 150	Detergents	R375
25 Car washes	R1 625	Motorcycle parts	…
		Water	R850,25
		Other	R1 500
TOTAL INCOME	…	TOTAL EXPENSES	R7 927,86

1.1.1 Calculate David's total income for the month. (2)

1.1.2 Determine the amount charged per motorcycle service. (2)

1.1.3 Calculate the amount spent on motorcycle parts. (2)

1.1.4 Calculate what percentage of total expenses David spends on water. (3)

1.2 TABLE 1 below shows the tollgate tariffs for the N1 road.

TABLE 1: N1 TOLLGATE TARIFFS

Tollgate (N1)		Class 1	Class 2	Class 3	Class 4
Huguenot	■ Main line	R44,50	R123,00	R193,00	R313,00
Vaal Plaza	■ Main line	R74,50	R140,00	R169,00	R225,00
Grasmere	■ Main line	R22,50	R67,00	R78,00	R103,00
	■ Ramp (N)	R11,50	R33,00	R39,00	R51,00
	■ Ramp (S)	R11,50	R33,00	R39,00	R51,00
Verkeerdevlei	– Main line	R64,00	R128,00	R193,00	R271,00

VEHICLE CLASS APPLICABLE TO CONVENTIONAL TOLL PLAZAS

CLASS 1 ALL LIGHT VEHICLES

HEAVY VEHICLES

CLASS 2 2 AXLES

CLASS 3 3 AND 4 AXLES

CLASS 4 5 OR MORE AXLES

CLECLASS APPLIC [Source: aa.co.za/toll-tariffs]

1.2.1 Determine the cost of travelling by car, towing a caravan, at the Grasmere Main line tollgate. (2)

1.2.2 If the cost of travelling by motorbike was R74,50, write down which tollgate was used. (2)

1.2.3 Write down the tariff for the Vaal Plaza, class 2 rates, as a simplified ratio to the class 4 rates of the same plaza. (2)

1.2.4 Calculate the cost of a return trip for a class 4 vehicle going through the Verkeerdevlei tollgate. (2)

Copyright reserved Please turn over1.3 The graph given below shows the favourite movie genres of a group of 220 students.
Use the above information to answer the questions that follow.
1.3.1 Name the type of graph represented above. (2)
1.3.2 Calculate the percentage of students that prefer comedy. (2)
1.3.3 Determine the number of students that prefer romance movies. (2)
[23]
Romance
24%
Thriller
11%
ComedyAction
21%Science Fiction
6%FA VOURITE MOVIE GENRES

Copyright reserved Please turn overQUESTION 2
2.1 The graph given below shows South Africa's CPI inflation rate for the period April 2022
to February 2023.
[Source: tradingeconomics. com/statssa]
Use the above information to answer the questions that follow.
2.1.1 In December 2022 the price of a loaf of bread was R12,47. Calculate the price of a loaf of bread in February 2023. (5)
2.1.2 Did the price of goods increase or decrease from July 2022 to September 2022? Explain your answer. (2)
2.2 The invoice given below is for the installation of a water tank.
WATER TANKS 4 YOU
Bill to: Mr T. Rock
1112 Flamingo Drive Invoice number: 123 765 987
Invoice date: 22/04/2023
Description Unit Price Amount
1 x 5 000 ■ Water tank R5 499,00 R5 499,00
3 x Filters R349,00 A
1 x Water pump R1 699,00 R1 699,00
Labour 7,5 hours x … R6 500,00
SUBTOTAL …
15% VAT …
INVOICE TOTAL …
*All payments due within 15 days of receipt of invoice. 5,9%6,5%7,4%7,8%7,6% 7,5%
7,6%7,4%7,2%6,9% 7,0%
0,0%1,0%2,0%3,0%4,0%5,0%6,0%7,0%8,0%9,0%South Africa CPI Inflation

Use the above information to answer the questions that follow.

2.2.1 Give the day and month this invoice must be paid. (2)

2.2.2 Determine the total cost (rounded to the nearest thousand) of the filters. (3)

2.2.3 Calculate the cost per hour for labour. (2)

2.2.4 Calculate the total amount due, including VAT, for the above invoice. (3)

2.2.5 The price of the water tank is due to increase in the next month. The new price is
expected to be R5 795,00. Mr Rock says that this increase is more than 5%.
Verify, showing
ALL calculations, whether this statement is valid or not.

You may use the formula:

Percentage change = Difference in cost
Original cost ×100%

(4)

2.3 Given in the table below are the electricity tariffs for domestic households.

BLOCKS ELECTRICITY RATE (c/kWh)
Block 1 0–600 kWh 229,00 c/kWh
Block 2 More than 600 kWh 278,46 c/kWh

Use the above information to answer the questions that follow.

Mr Rock and his family used a total of 712 kWh of electricity. Calculate the amount, in Rands, that he will pay for their electricity usage this month. (4)
[25]

QUESTION 3

TABLE 2 below lists the winnings, endorsements and total net worth of the nine highest paid athletes in 2022. All values in the table are given in millions.

TABLE 2: TOP NINE HIGHEST PAID ATHLETES IN 2022
Name Winnings
(in millions) Endorsements
(in millions) Total Net Worth
(in millions)
Lionel Messi (soccer) $75 $55 $130
Le Bron James (basketball) $41,2 $80 $121,2
Cristiano Ronaldo (soccer) $60 $55 $115
Neymar (soccer) $70 $25 $95
Stephan Curry (soccer) $45,8 $47 $92,8
Kevin Durant (soccer) $42,1 $50 $92,1
Roger Federer (tennis) --- $90 ----
Canelo Alvarez (boxing) $85 $5 $90
Tom Brady (football) $31,9 $52 $83,9
TOTAL: B

[Source: Wikipedia.org]

Use the above information to answer the questions that follow.

3.1 Write the winnings of Stephan Curry in words. (2)

3.2 Determine the median value of the endorsements of the nine players. (3)

3.3 Calculate the value of B, the total net worth of the nine players, if the total net worth of Roger Federer is $30,5 million dollars less than that of Le Bron James. (4)

3.4 Calculate the mean winnings for 2022. (3)

3.5 Express the endorsement amount of Cristiano Ronaldo as a percentage of his total net worth. (2)

3.6 Write down the winnings of Neymar as a simplified ratio to the total net worth of Canelo Alvarez. (2)

3.7 Determine the probability as a decimal fraction (rounded to THREE decimal places), of

randomly selecting a player from the table a bove that plays any sport except soccer. (3)
[19]

QUESTION 4

4.1 The choir committee of a high school have started a small business to raise funds for their upcoming tour. They make and sell curry bunnies with assistance from their parents. They pay their parents R450 each month to cove r costs for water and electricity. It costs them R10,00 to make one curry bunny.

TABLE 3 below shows their income and expenses for one month.

TABLE 3: CHOIR COMMITTEES MONTHLY INCOME AND EXPENSES

Number of curry bunnies	0	10	20	50	100	150	200
Expenses (in Rands)	450	550	650	950	1 450	1 950	2 450
Income (in Rands)	0	250	500	1 250	2 500	3 750	5 000

Use the above information to answer the questions that follow.

4.1.1 Identify the dependent and independent variables in the above context. (2)

4.1.2 Determine the selling price of ONE curry bunny. (2)

4.1.3 Write down the formula they will use to determine their total expenses. (2)

4.1.4 The graph drawn on the ANSWER SHEET shows the total income. On the same set of axes, draw a graph showing the TOTAL EXPENSES for the number of curry bunnies made in one month. (3)

4.1.5 The choir committee claims that if they sell 200 curry bunnies every month for four months, they will make more than R10 000 profit.

Verify, showing all calculations, whether their statement is valid or not. (4)

4.2 The frequency table below shows the number of motorists buying petrol on a specific day.

Time Interval	Frequency	Cumulative Frequency
06:00 –08:59	12	12
09:00 –11:59	8	20
12:00 –14:59	13	C
D	17	50
18:00 –20:59	23	73
21:00 – 23:59	4	77

Use the above information to answer the questions that follow.

4.2.1 Determine the value of C. (2)

4.2.2 Give the time interval at D. (2)

4.2.3 Write down the modal time interval. (2)

4.2.4 Each motorist buys an average of 6,5■ of petrol on this day. If the price of petrol is R21,92 per litre, determine the total cost for the petrol bought on this day. (3)

4.2.5 What is the probability, as a fraction, of a motorist buying petrol before 12:00 on this day? (2)

4.3 Sethu wants to save money to buy her mother a gift for her 50th birthday, which is in four years' time. She decides to look at two banks, both offering an interest rate of 8,5% p.a.

She has R1 000 that she can save. The table below shows the amount th at she will have saved at each bank annually.

	YEAR 1	YEAR 2	YEAR 3	YEAR 4
BANK A	R1 085	R1 177,23	R1 277,29	---
BANK B	R1 085	R1 170	R1 255	R1 340

Use the above information to answer the questions that follow.

4.3.1 Which bank represents simple interest? Give a reason for your answer. (2)

4.3.2 Sethu claims that if she invests at BANK A, she will have saved R45,86 more than she would at BANK B, after four years.

Verify, showing all calculations, whether her statement is valid or not. (4)

4.3.3 Sethu would like to order her mother a clock from Switzerland. It costs 68,49 Euros. The exchange rate is 1 Euro = R19,83.

If she decides to invest with Bank A, will she have saved enough after four years to afford the gift? (3)

[33]

TOTAL: 100

ANSWER SHEET

NAME OF LEARNER: …………………………………… GRADE: ……

QUESTION 4.1.4

04509001350180022502700315036004050450049505400
0 50 100 150 200250Amount in Rands
Number of curry bunniesINCOME AND EXPENSE OF CURRY BUNNY SALES
INCOME

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2023

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including an answer sheet .

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ANSWER SHEET to answer QUESTION 4.1.4.

3. Number the answers correctly according to the numbering system used in this question paper.

4. Diagrams are not necessarily drawn to scale, unless stated otherwise.

5. Round off ALL final answers according to the context used, unless stated otherwise.

6. Indicate units of measurement, where applicable.

7. Start EACH question on a NEW page.

8. Show ALL calculations clearly.

9. Write neatly and legibly.

QUESTION 1

1.1 David runs a small motorcycle service business from home. Given below is the budget drawn up by him for the new month.

INCOME	AMOUNT	EXPENSES	AMOUNT
26 Motorcycle services	R33 150	Detergents	R375
25 Car washes	R1 625	Motorcycle parts	…
		Water	R850,25
		Other	R1 500
TOTAL INCOME	…	TOTAL EXPENSES	R7 927,86

1.1.1 Calculate David’s total income for the month. (2)

1.1.2 Determine the amount charged per motorcycle service. (2)

1.1.3 Calculate the amount spent on motorcycle parts. (2)

1.1.4 Calculate what percentage of total expenses David spends on water. (3)

1.2 TABLE 1 below shows the tollgate tariffs for the N1 road.

TABLE 1: N1 TOLLGATE TARIFFS

Tollgate (N1)		Class 1	Class 2	Class 3	Class 4
Huguenot	■ Main line	R44,50	R123,00	R193,00	R313,00
Vaal Plaza	■ Main line	R74,50	R140,00	R169,00	R225,00
Grasmere	■ Main line	R22,50	R67,00	R78,00	R103,00
	■ Ramp (N)	R11,50	R33,00	R39,00	R51,00
	■ Ramp (S)	R11,50	R33,00	R39,00	R51,00
Verkeerdevlei	– Main line	R64,00	R128,00	R193,00	R271,00

VEHICLE CLASS APPLICABLE TO CONVENTIONAL TOLL PLAZAS
CLASS 1 ALL LIGHT VEHICLES

HEAVY VEHICLES
CLASS 2 2 AXLES

CLASS 3 3 AND 4 AXLES

CLASS 4 5 OR MORE AXLES

CLECLASS APPLIC [Source: aa.co.za/toll -tariffs]

1.2.1 Determine the cost of travelling by car, towing a caravan, at the Grasmere Main line tollgate. (2)

1.2.2 If the cost of travelling by motorbike was R74,50, write down which tollgate was used. (2)

1.2.3 Write down the tariff for the Vaal Plaza , class 2 rates, as a simplified ratio to the class 4 rates of the same plaza. (2)

1.2.4 Calculate the cost of a return trip for a class 4 vehicle going through the Verkeerdevlei tollgate. (2)

1.3 The graph given below shows the favourite movie genres of a group of 220 students.

Use the above information to answer the questions that follow.

1.3.1 Name the type of graph represented above. (2)

1.3.2 Calculate the percentage of students that prefer comedy. (2)

1.3.3 Determine the number of students that prefer romance movies. (2)
[23]

Romance
24%
Thriller
11%
ComedyAction
21%Science Fiction
6%FAVOURITE MOVIE GENRES

QUESTION 2

2.1 The graph given below shows South Africa’s CPI inflation rate for the period April 2022 to February 2023.

[Source: tradingeconomics.com/statssa]

Use the above information to answer the questions that follow.

2.1.1 In December 2022 the price of a loaf of bread was R12,47. Calculate the price of a loaf of bread in February 2023. (5)

2.1.2 Did the price of goods increase or decrease from July 2022 to September 2022? Explain your answer. (2)

2.2 The invoice given below is for the installation of a water tank.

WATER TANKS 4 YOU
Bill to: Mr T. Rock
1112 Flamingo Drive   Invoice number: 123 765 987
Invoice date: 22/04/2023

Description	Unit Price	Amount
1 x 5 000 ■ Water tank	R5 499,00	R5 499,00
3 x Filters	R349,00	A
1 x Water pump	R1 699,00	R1 699,00
Labour	7,5 hours x …	R6 500,00
SUBTOTAL		…
15% VAT		…
INVOICE TOTAL		…

*All payments due within 15 days of receipt of invoice.

5,9%6,5%7,4%7,8%7,6% 7,5% 7,6%7,4%7,2%6,9% 7,0%
0,0%1,0%2,0%3,0%4,0%5,0%6,0%7,0%8,0%9,0%South Africa CPI Inflation

Use the above information to answer the questions that follow.

2.2.1 Give the day and month this invoice must be paid. (2)

2.2.2 Determine the total cost (rounded to the nearest thousand) of the filters. (3)

2.2.3 Calculate the cost per hour for labour. (2)

2.2.4 Calculate the total amount due, including VAT, for the above invoice. (3)

2.2.5 The price of the water tank is due to increase in the next month. The new price is expected to be R5 795,00. Mr Rock says that this increase is more than 5%.

Verify, showing ALL calculations, whether this statement is valid or not.

You may use the formula:

$$\text{Percentage change} = \frac{\text{Difference in cost}}{\text{Original cost}} \times 100\%$$

(4)

2.3 Given in the table below are the electricity tariffs for domestic households .

BLOCKS	ELECTRICITY	RATE (c/kWh)
Block 1	0–600 kWh	229,00 c/kWh
Block 2	More than 600 kWh	278,46 c/kWh

Use the above information to answer the questions that follow.

Mr Rock and his family used a total of 712 kWh of electricity. Calculate the amount, in Rands, that he will pay for their electricity usage this month. (4)

[25]

QUESTION 3

TABLE 2 below lists the winnings, endorsements and total net worth of the nine highest paid athletes in 2022. All values in the table are given in millions.

TABLE 2: TOP NINE HIGHEST PAID ATHLETES IN 2022

Name	Winnings (in millions)	Endorsements (in millions)	Total Net Worth (in millions)
Lionel Messi (soccer)	$75	$55	$130
Le Bron James (basketball)	$41,2	$80	$121,2
Cristiano Ronaldo (soccer)	$60	$55	$115
Neymar (soccer)	$70	$25	$95
Stephan Curry (soccer)	$45,8	$47	$92,8
Kevin Durant (soccer)	$42,1	$50	$92,1
Roger Federer (tennis)	---	$90	----
Canelo Alvarez (boxing)	$85	$5	$90
Tom Brady (football)	$31,9	$52	$83,9
TOTAL:			B

[Source: Wikipedia.org]

Use the above information to answer the questions that follow.

3.1 Write the winnings of Stephan Curry in words. (2)

3.2 Determine the median value of the endorsements of the nine players. (3)

3.3 Calculate the value of B, the total net worth of the nine players, if the total net worth of Roger Federer is $30,5 million dollars less than that of Le Bron James. (4)

3.4 Calculate the mean winnings for 2022. (3)

3.5 Express the endorsement amount of Cristiano Ronaldo as a percentage of his total net worth. (2)

3.6 Write down the winnings of Neymar as a simplified ratio to the total net worth of Canelo Alvarez. (2)

3.7 Determine the probability as a decimal fraction (rounded to THREE decimal places), of randomly selecting a player from the table above that plays any sport except soccer. (3)

[19]

QUESTION 4

4.1 The choir committee of a high school have started a small business to raise funds for their upcoming tour. They make and sell curry bunnies with assistance from their parents.

They pay their parents R450 each month to cover costs for water and electricity. It costs them R10,00 to make one curry bunny.

TABLE 3 below shows their income and expenses for one month.

TABLE 3: CHOIR COMMITTEES MONTHLY INCOME AND EXPENSES

Number of curry bunnies	0	10	20	50	100	150	200
Expenses (in Rands)	450	550	650	950	1 450	1 950	2 450
Income (in Rands)	0	250	500	1 250	2 500	3 750	5 000

Use the above information to answer the questions that follow.

4.1.1 Identify the dependent and independent variables in the above context. (2)

4.1.2 Determine the selling price of ONE curry bunny. (2)

4.1.3 Write down the formula they will use to determine their total expenses. (2)

4.1.4 The graph drawn on the ANSWER SHEET shows the total income. On the same set of axes, draw a graph showing the TOTAL EXPENSES for the number of curry bunnies made in one month. (3)

4.1.5 The choir committee claims that if they sell 200 curry bunnies every month for four months, they will make more than R10 000 profit.

Verify, showing all calculations , whether their statement is valid or not. (4)

4.2 The frequency table below shows the number of motorists buying petrol on a specific day.

Time Interval	Frequency	Cumulative Frequency
06:00 –08:59	12	12
09:00 –11:59	8	20
12:00 –14:59	13	C
D	17	50
18:00 –20:59	23	73
21:00 – 23:59	4	77

Use the above information to answer the questions that follow.

4.2.1 Determine the value of C. (2)

4.2.2 Give the time interval at D. (2)

4.2.3 Write down the modal time interval. (2)

4.2.4 Each motorist buys an average of 6,5■ of petrol on this day. If the price of petrol is R21,92 per litre, determine the total cost for the petrol bought on this day. (3)

4.2.5 What is the probability, as a fraction, of a motorist buying petrol before 12:00 on this day? (2)

4.3 Sethu wants to save money to buy her mother a gift for her 50th birthday, which is in four years' time. She decides to look at two banks, both offering an interest rate of 8,5% p.a.

She has R1 000 that she can save. The table below shows the amount that she will have saved at each bank annually.

	YEAR 1	YEAR 2	YEAR 3	YEAR 4
BANK A	R1 085	R1 177,23	R1 277,29	---
BANK B	R1 085	R1 170	R1 255	R1 340

Use the above information to answer the questions that follow.

4.3.1 Which bank represents simple interest? Give a reason for your answer. (2)

4.3.2 Sethu claims that if she invests at BANK A, she will have saved R45,86 more than
she would at BANK B, after four years.

Verify , showing all calculations , whether her statement is valid or not. (4)

4.3.3 Sethu would like to order her mother a clock from Switzerland. It costs 68,49 Euros. The exchange rate is 1 Euro = R19,83.

If she decides to invest with Bank A, will she have saved enough after four years
to afford the gift? (3)
[33]

TOTAL: 100

ANSWER SHEET

NAME OF LEARNER: …………………………………… GRADE: ……

QUESTION 4.1.4

045090013501800225027003150360040504500495054000
0 50 100 150 200 250Amount in Rands
Number of curry bunniesINCOME AND EXPENSE OF CURRY BUNNY SALES
INCOME

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2023

MATHEMATICAL LITERACY P2
(DEAF)

MARKS: 100

TIME: 2 hours

This question paper has 11 pages , with an addendum with 2 annexures.

INSTRUCTIONS AND INFORMATION

1. This question paper has FOUR questions .

2. Use the ANNEXURE S in the ADDENDUM for:

- ANNEXURE A for QUESTION 2 .1
- ANNEXURE B for QUESTION 4.1

3. Answer ALL the questions .

4. Number the answers the same as the numbers on the question paper .

5. Diagrams are NOT drawn to scale .
Some questions will tell you to use the scale .

6. Round off ALL final answers to fit with the content of the question.

7. Write units where needed.

8. Start EACH question on a NEW page .

9. Show ALL calculations .

10. Write neatly .
Your work should be easy to read .

QUESTION 1

1.1 A young entrepreneur stocks sheet rolls for securing products on pallets .
These rolls are suitable for wrapping goods .
They are sold in 200 m and 300 m rolls .

D = 450 mm

[Source: www.supplywise.co.za]

Use the information . Answer the questions .

1.1.1 The circumference is 3,142 times more than the diameter .
Calculate the circumference of the roll in millimetre (mm). (2)

1.1.2 Calculate the cost of the pallet sheet roll per meter for Option B. (2)

1.1.3 Write the cost for Option B to Option A in simplified ratio format . (2)

1.1.4 Determine the radius in centimetre (cm). (3)

Option A:
300 m sheet roll cost R390,00

Option B:
200 m sheet roll cost R2 90,00

Diameter of roll (D) = 450 mm

1.2 David and his family travelled from Port -Elizabeth (now Gqeberha) to Durban for the school holidays.
The map below shows the journey the family undertook .

Use the information . Answer the questions .

1.2.1 Name TWO national roads indicated (shown) on the map . (2)

1.2.2 Identify the type of map shown. (2)

1.2.3 Determine the actual (real) distance between Port -Elizabeth and Durban in metres (m). (3)

1.2.4 David quickly visited his cousin in Margate.
Just after Umtata (Mthatha) he entered (took) the R61 to Margate .

(a) Write the name of ONE town he passed between Umtata (Mthatha) and Margate . (2)

(b) Calculate the total distance , in kilometres (km) that he travelled from Port St. Johns to Margate . (3)

1.2.5 Name TWO provincial roads on the map . (2)
[23]

QUESTION 2

2.1 A group of four university friends plan to watch the rugby game between the Springboks and All Blacks in Durban .
They plan to travel by car and share the cost of the trip.

On ANNEXURE A is a map of South Africa .

Use ANNEXURE A of the addendum . Answer the questions .

2.1.1 Identify the type of scale on the map . (2)

2.1.2 What is the general direction from Umtata (Mthatha) to Cape Town ? (2)

2.1.3 Durban and Upington are 9,6 cm apart on the map .
Mr Antonie claims that the distance between Durban and Upington is 972 km .

Say if Mr Antonie is correct . Use the bar scale method . (4)

2.2 Amos and his friends took exactly 17,25 hours to drive 1 635 km from Cape Town to Durban for the rugby game between the Springboks and the All Blacks .

2.2.1 Determine Amos’s average speed for the trip in km/h .

Use the formula : Speed = Distance ÷ Time (3)

2.2.2 The petrol consumption (use) of the car is 1 litre per 1 2,5 km.

(a) Amos claimed that if the petrol consumption (use) was 0,80 litre per 10 km , the car would use less petrol .

Say if Amos is correct . Show ALL calculations. (5)

(b) The petrol price is R24,75 per litre .
Calculate the cost of petrol to drive from Cape Town to Durban . (2)
[18]

QUESTION 3

3.1 Uyathanda Home Industry specialises in baking and selling cakes of all types .
The recipe of a cake is shown below.
Study the recipe .
Answer the questions .

INGREDIENTS SOUR CREAM CHOCOLATE
CAKE
• 3
4 cups (250 g) flour
• 13
4 cups (360 g) sugar
• 3
4 cup (90 g) unsweetened cocoa
powder
• 2 teaspoons baking powder
• 1 teaspoon kosher salt
• 2 large eggs
• 1 cup sour cream
• 3
4 cup canola oil
• 2 teaspoons vanilla extract
• 1 cup strong piping (very) hot coffee
• 1 mixture (11
2 cups) chocolate or
cream cheese frosting
• Preparation time: 20 min utes
• Baking time: 55 minutes
• Total time: 75 minutes

Other information:
• Preheat the oven to 320 ■

NOTE:
• An order was received for 90 people who will be attending an event .
• Each person must get one slice of cake .
• 12 slices can be cut from one cake .
• One cake weighs 900 g .
• The amount of energy in 100 g of cake is 400 calories .

3.1.1 Write down the mass of one cake in kilograms. (2)

3.1.2 Determine the mass of one slice of cake in grams. (2)

3.1.3 Calculate the number of calories in one slice of cake. (3)

3.1.4 Convert (Change) total time in minutes to hours. (2)

3.1.5 Determine the number of cakes that should be baked for the number of guests at the event. (4)

3.2 3.2.1 How many cups of flour is required (needed) if eight (8) cakes must be baked? (2)

3.2.2 The cost of 240 g of unsweetened cocoa powder is R62,75.
Determine how much money will be needed for unsweetened cocoa
powder for eight (8) cakes. (4)

3.2.3 Calculate the temperature of 320 °F in degrees Celsius.

Use the formula : ■ = (■ – 32) ÷ 1,8 (3)

3.2.4 If a sour cream chocolate cake is placed in the oven at 09h03, at what time will the cake be ready? (2)

3.3 Mr Sihle owns a company which focus primarily (mainly) on producing cylindrical
metal cans for the unsweetened cocoa powder.
The volume of the can is 546,10 cm3.
The height of the can's label is 80% of the height of the can.

Use the diagram. Answer the questions.

3.3.1 Determine the height of the can in centimetre (cm). (4)

3.3.2 Mr Sihle stated that the height of the can is 1,5 cm more than the height of
the label.
Say if Mr Sihle is correct. Show ALL calculations. (4)
[32]

NOTE:
You may use the formula below.
Volume = 3,142 × r2 × height Height of the can
Diameter = 86 mm

QUESTION 4

4.1 Mrs Aretha Smith has a floor plan for the new house she wants to build.
Use ANNEXURE B. It shows an image of the floor plan of this house.

Use ANNEXURE B. Answer the questions.

4.1.1 Explain the term ‘floor plan ’. (2)

4.1.2 How many windows are there on the east wall of the house ? (2)

4.1.3 Use the number scale given .
Determine the actual total length of all the outside walls of the house on
the Northern and Eastern sides .
Give your final answer in metres . (7)

4.1.4 The area of the porch is 19,38 m2, and the length is 10,2 m.
Mrs Smith says that the width is six times less than the length .
Say if Mrs Smith is correct . Show ALL calculations. (4)

Use the formula : Area = length × breadth

4.2 Mr Smith surprise s his wife and gives her a rare (not easy to find) lucky coin
on her
birthday .
The coin has a square cut out of the middle as shown .

NOTE:
Length of one side of the square = 0,9 cm
Diameter of circle = 3,3 cm
Volume of the coin = 1,47 cm3

FORMULAE:
Area of circle = ■× r2 ; where ■ = 3,142
Area of square = side × side

Density = Mass
Volume

Use the information . Answer the questions .

4.2.1 Calculate the area of the coin in cm2.
Round your answer off to ONE decimal place . (5)

4.2.2 The density of gold is 19,30 g / cm3.
Calculate the mass of the coin in grams . (2)
Round your answer off to ONE decimal place .

[Source : www.walmart.com]

4.3 A box contains (holds) 12 gold coins , 2 silver coins and 2 bronze coins .

4.3.1 Determine the probability of selecting a gold coin in decimal format . (3)

4.3.2 Use your answer in QUESTION 4.3.1 .
Explain the probability of the event in words . (2)
[27]

TOTAL: 100

ANNEXURE A

QUESTION 2.1

Below is a map of South Africa .

[Source: http://www.lonelyplanet.com/maps/Africa/south -africa/map of south -africa.jpg]

ANNEXUR E B

QUESTION 4.1

Floor plan of Mrs Smith’s house :

Scale 1 : 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2023

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 11 pages, including an addendum with 2 annexures.

imlit2

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.

2. Use the ANNEXURE in the ADDENDUM to answer the following questions.

x ANNEXURE A for QUESTION 2.1
x ANNEXURE B for QUESTION 4.1

3. Answer ALL the questions.

4. Number the questions correctly according to the numbering system used in this question paper.

5. Diagrams are NOT necessarily drawn to scale.

6. Round off ALL the final answers appropriately according to the context used, unless stated otherwise.

7. Indicate units of measurement, where applicable.

8. Start EACH question on a NEW page.

9. Show ALL calculations clearly.

10. Write neatly and legibly.

QUESTION 1

1.1 A young entrepreneur stocks sheet rolls for securing products on pallets. These rolls are suitable for wrapping goods. They are sold in 200 m and 300 m rolls.

D = 450 mm

[Source: www.supplywise.co.za]

Use the information above to answer the questions that follow.

1.1.1 The circumference is 3,142 times more than the diameter. Calculate the circumference of the roll in mm. (2)

1.1.2 Calculate the cost of the pallet sheet roll per meter for Option B. (2)

1.1.3 Write the cost for Option B to Option A in simplified ratio format. (2)

1.1.4 Determine the radius in cm. (3)

Option A: 300 m
sheet roll cost R 390,00
Option B:

200 m sheet roll cost R 290,00

Diameter of roll (D) = 450 mm

1.2 David and his family travelled from Port -Elizabeth (now Gqeberha) to Durban for the school holidays. The map below shows the journey the family undertook.

Use the information above to answer the questions that follow.

1.2.1 Name TWO national roads indicated on the map. (2)

1.2.2 Identify the type of map shown. (2)

1.2.3 Determine the actual distance between Port-Elizabeth and Durban in metres. (3)

1.2.4 David quickly visited his cousin in Margate. Just after Umtata he entered the R61 to Margate.

(a) Write down the name of ONE town he passed between Umtata and Margate. (2)

(b) Calculate the total distance, in km that he travelled from Port St. Johns to Margate. (3)

1.2.5 Name TWO provincial roads on the map. (2)
[23]

QUESTION 2

2.1 A group of four university friends plan to watch the rugby game between the Springboks and All Blacks in Durban . They plan to travel by car and share the cost of the trip.

On ANNEXURE A is a map of South Africa.

Use ANNEXURE A of the addendum to answer the questions that follow.

2.1.1 Identify the type of scale on the map. (2)

2.1.2 What is the general direction from Umtata to Cape Town? (2)

2.1.3 Durban and Upington are 9,6 cm apart on the map. Mr Antonie claims that the distance between Durban and Upington is 972 km.

Verify if his statement is correct by using the bar scale method. (4)

2.2 Amos and his friends took exactly 17,25 hours to drive 1 635 km from Cape Town to Durban for the rugby game between the Springboks and the All Blacks.

2.2.1 Determine Amos's average speed for the trip in km/h.

You may use the following formula: Speed = Distance ÷ Time (3)

2.2.2 The petrol consumption of the car is 1 litre per 12,5 km.

(a) Amos claimed that if the petrol consumption was 0,80 litre per 10 km, the car would use less petrol. Verify with the necessary calculation if his statement is valid. (5)

(b) Calculate the cost of petrol to drive from Cape Town to Durban. The petrol price is R24,75 per litre. (2)
[18]

QUESTION 3

3.1 Uyathanda Home Industry specialises in baking and selling cakes of all types. The recipe of the cake is shown below. Study the recipe and answer the questions that follow .

INGREDIENTS SOUR CREAM CHOCOLATE CAKE
x 3
4 cups (250 g) flour
x 13
4 cups (360 g) sugar
x 3
4 cup (90 g) unsweetened cocoa powder
x 2 teaspoons baking powder
x 1 teaspoon kosher salt

x 2 large eggs
x 1 cup sour cream
x 3
4 cup canola oil
x 2 teaspoons vanilla extract
x 1 cup strong piping hot coffee
x 1 mixture (11
2 cups) chocolate or
cream cheese frosting
x Preparation time: 20 minutes
x Cook time: 55 minutes
x Total time: 75 minutes

Other information:
x Preheat the oven to 320 ■

NOTE:
x An order was received for 90 people who will be attending an event.
x Each person must get one slice of cake.
x 12 slices can be cut from one cake.
x One cake weighs 900 g.
x The amount of energy in 100 g of cake is 400 calories.

3.1.1 Write down the mass of one cake in kilograms. (2)

3.1.2 Determine the mass of one slice of cake in grams. (2)

3.1.3 Calculate the number of calories in one slice of cake. (3)

3.1.4 Convert total time in minutes to hours. (2)

3.1.5 Determine the number of cakes that should be baked for the number of guests at the event. (4)

Copyright reserved Please turn over3.2 3.2.1 How many cups of flour is required if eight cakes must be baked? (2)
3.2.2 The cost of 240 g of unsweetened cocoa powder is R62,75. Determine how much money will be needed for unsweetened cocoa powder for 8 cakes. (4)
3.2.3 Calculate the temperature of 320 °F in degrees Celsius.
Use the formula: ■=(■– 32) ÷1,8 (3)
3.2.4 If a sour cream chocolate cake is plac ed in the oven at 09h03, at what time will the cake be ready? (2)
3.3 Mr Sihle owns a company which focus primarily on producing cylindrical metal cans for the unsweetened cocoa powder. The volume of the can is 546,10 cm3. The height of the can's label is 80 %of the height of the can.
Refer to the diagram below and answer the questions that follows.
3.3.1 Determine the height of the can in cm. (4)
3.3.2 Mr Sihle stated that the height of the can is 1,5 cm more than the height of the label. Verify by means of a calculati on whether his statement is valid. (4)
[32]
NOTE:
You may use the formula below .
Volume =3,142 ×r2×heightHeight of the can
Diameter =86 mm

Copyright reserved Please turn overQUESTION 4
4.1 Mrs Aretha Smith has a floor plan for the new house she wants to build. Refer to

ANNEXURE B which shows an image of the floor plan of this house.
Use ANNEXURE B to answer the questions that follow.

4.1.1 Explain the term ‘floor plan ’. (2)

4.1.2 How many windows are there on the east wall of the house? (2)

4.1.3 Use the number scale give n and determine the actual total length of all the outside walls of the house on the Northern and Eastern sides . Give your final answer in metres. (7)

4.1.4 The area of the porch is 19,38 m2, and the length is 10,2 m. Mrs Smith stated that the width is six times less than the length.
Verify with the necessary calculations if her statement is valid. (4)
You may use the formula: Area =length ×breadth

4.2 Mr Smith surprises his wife and gives he r a rare lucky coin on her birthday. The coin has a square cut out of the middle as shown in the photo below.

NOTE:
Length of one side of the square =0,9 cm
Diameter of circle =3,3 cm
Volume of the coin =1,47 cm3

FORMULAE:
Area of circle =■×r2 ; where ■=3,142
Area of square =side ×side
Density =Mass
Volume

Use the information above to answer the questions that follow.

4.2.1 Calculate the area of the coin in cm2. Round your answer off to ONE decimal place. (5)

4.2.2 The density of gold is 19,30 g / cm3, calculate the mass of the coin in grams. (2)
Round your answer off to ONE decimal place. [Source: www.walmart.com]

4.3 A box contains 12 gold coins, 2 silver coins and 2 bronze coins.

4.3.1 Determine the probability of selecting a gold coin in decimal format. (3)

4.3.2 Refer to your answer in QUESTION 4.3.1. Explain the probability of the event in words. (2)
[27]

TOTAL: 100

ANNEXURE A

QUESTION 2.1

Below is a map of South Africa .

[Source: http://www.lonelyplanet.com/maps/Afr ica/south-africa/map of south-africa.jpg]

ANNEXURE B

QUESTION 4.1

Floor plan of Mrs Smith’s house:

Scale 1 : 100

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2023

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 11 pages , including an addendum with 2 annexures.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.

2. Use the ANNEXURE in the ADDENDUM to answer the following questions .

- ANNEXURE A for QUESTION 2 .1
- ANNEXURE B for QUESTION 4.1

3. Answer ALL the questions.

4. Number the questions correctly according to the numbering system used in this question paper.

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7. Indicate units of measurement, where applicable.

8. Start EACH question on a NEW page.

9. Show ALL calculations clearly.

10. Write neatly and legibly.

QUESTION 1

1.1 A young entrepreneur stocks sheet rolls for securing products on pallets. These rolls are suitable for wrapping goods. They are sold in 200 m and 300 m rolls.

D = 450 mm

[Source: www.supplywise.co.za]

Use the information above to answer the questions that follow.

1.1.1 The circumference is 3,142 times more than the diameter. Calculate the circumference of the roll in mm. (2)

1.1.2 Calculate the cost of the pallet sheet roll per meter for Option B. (2)

1.1.3 Write the cost for Option B to Option A in simplified ratio format . (2)

1.1.4 Determine the radius in cm . (3)

Option A:
300 m sheet roll cost R390,00

Option B:
200 m sheet roll cost R2 90,00

Diameter of roll (D) = 450 mm

1.2 David and his family travelled from Port -Elizabeth (now Gqeberha) to Durban for the school holidays. The map below shows the journey the family undertook.

Use the information above to answer the questions that follow .

1.2.1 Name TWO national road s indicated on the map. (2)

1.2.2 Identify the type of map shown. (2)

1.2.3 Determine the actual distance between Port -Elizabeth and Durban in metres. (3)

1.2.4 David quickly visited his cousin in Margate. Just after Um tata he entered the R61 to Margate .

(a) Write down the name of ONE town he passed between Umtata and Margate. (2)

(b) Calculate the total distance, in km that he travelled from Port St. Johns to Margate. (3)

1.2.5 Name TWO provincial roads on the map . (2)
[23]

QUESTION 2

2.1 A group of four univer sity friends plan to watch the rugby game between the Springboks and All Blacks in Durban . They plan to travel by car and share the cost of the trip.

On ANNEXURE A is a m ap of South Africa .

Use ANNEXURE A of the addendum to answer the questions that follow.

2.1.1 Identify the type of scale on the map . (2)

2.1.2 What is the general direction from Umtata to Cape Town ? (2)

2.1.3 Durban and Upington are 9,6 cm apart on the map. Mr Antonie claim s that the distance between Durban and Upington is 972 km.

Verify if his statement is correct by using the bar scale method. (4)

2.2 Amos and his friends took exactly 1 7,25 hours to drive 1 635 km from Cape Town to Durban for the rugby game between the Springboks and the All Blacks .

2.2.1 Determine Amos's average speed for the trip in km/h.

You may use the following formula: Speed = Distance ÷ Time (3)

2.2.2 The petrol consumption of the car is 1 litre per 1 2,5 km.

(a) Amos claimed that if the petrol consumption was 0,80 litre per 10 km , the car would use less petrol. Verify with the necessary calculation if his statement is valid . (5)

(b) Calculate the cost of petrol to drive from Cape Town to Durban . The petrol price is R2 4,75 per litre. (2)
[18]

QUESTION 3

3.1 Uyathanda Home Industry specialises in baking and selling cakes of all types. The recipe of a cake is shown below. Study the recipe and answer the questions that follow .

INGREDIENTS SOUR CREAM CHOCOLATE CAKE
• 3
4 cups (250 g) flour
• 13
4 cups (360 g) sugar
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4 cup (90 g) unsweetened cocoa powder
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• 1 teaspoon kosher salt
• 2 large eggs
• 1 cup sour cream
• 3
4 cup canola oil
• 2 teaspoons vanilla extract
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• 1 mixture (11
2 cups) chocolate or cream cheese frosting
• Preparation time: 20 minutes
• Baking time: 55 minutes
• Total time: 75 minutes

Other information:
• Preheat the oven to 320 ■

NOTE:

- An order was received for 90 people who will be attending an event.
- Each person must get one slice of cake.
- 12 slices can be cut from one cake.
- One cake weighs 900 g.
- The amount of energy in 100 g of cake is 400 calories.

3.1.1 Write down the mass of one cake in kilogram s. (2)

3.1.2 Determine the mass of one slice of cake in grams . (2)

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3.1.4 Convert total time in minutes to hours . (2)

3.1.5 Determine the number of cakes that should be baked for the number of guests at the event . (4)

3.2 3.2.1 How many cups of flour is required if eight cakes must be baked? (2)

3.2.2 The cost of 240 g of unsweetened cocoa powder is R62,75. Determine how much money will be needed for unsweetened cocoa powder for 8 cakes . (4)

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Use the formula : ■ = (■ – 32) ÷ 1,8 (3)

3.2.4 If a sour cream chocolate cake is placed in the oven at 09h03, at what time will the cake be ready? (2)

3.3 Mr Sihle owns a company which focus primarily on producing cylindrical metal cans for the unsweetened cocoa powder. The volume of the can is 546,10 cm3. The height of the can's label is 80 % of the height of the can.

Refer to the diagram below and answer the questions that follows.

3.3.1 Determine the height of the can in cm. (4)

3.3.2 Mr Sihle stated that the height of the can is 1,5 cm more than the height of the label. Verify by means of a calculation whether his statement is valid. (4)

[32]

NOTE:
You may use the formula below .
Volume = 3,142 × r2 × height Height of the can
Diameter = 86 mm

QUESTION 4

4.1 Mrs Aretha Smith has a floor plan for the new house she wants to build . Refer to ANNEXURE B which shows an image of the floor plan of th is house .

Use ANNEXURE B to answer the questions that follow.

4.1.1 Explain the term ‘floor plan ’. (2)

4.1.2 How many windows are there on the east wall of the house? (2)

4.1.3 Use the number scale given and determine the actual total length of all the outside walls of the house on the Northern and Eastern sides . Give your final answer in metres . (7)

4.1.4 The area of the porch is 19,38 m2, and the length is 10,2 m. Mrs Smith stated that the width is six times less than the length.
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You may use the formula: Area = length × breadth

4.2 Mr Smith surprise s his wife and gives her a rare lucky coin on her birthday. The coin has a square cut out of the middle as shown in the photo below .

NOTE:
Length of one side of the square = 0,9 cm
Diameter of circle = 3,3 cm
Volume of the coin = 1,47 cm3

FORMULAE:
Area of circle = ■× r2 ; where ■ = 3,142
Area of square = side × side

Density = Mass
Volume

Use the information above to answer the questions that follow.

4.2.1 Calculate the area of the coin in cm2. Round your answer off to ONE decimal place . (5)

4.2.2 The density of gold is 19,30 g / cm3, calculate the mass of the coin in grams. (2)
Round your answer off to ONE decimal place.

[Source : www.walmart.com]

4.3 A box contains 12 gold coins, 2 silver coins and 2 bronze coins.

4.3.1 Determine the probability of selecting a gold coin in decimal format. (3)

4.3.2 Refer to your answer in QUESTION 4.3.1. Explain the probability of the event in words. (2)
[27]

TOTAL: 100

ANNEXURE A

QUESTION 2.1

Below is a map of South Africa.

[Source: http://www.lonelyplanet.com/maps/Africa/south -africa/map of south -africa.jpg]

ANNEXURE B

QUESTION 4.1

Floor plan of Mrs Smith's house:

Scale 1 : 100

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GRADE 11

NOVEMBER 2022

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 11 pages.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Number the answers correctly according to the numbering system used in this question

paper.

3. An approved calculator (non -programmable and non-graphical) may be used, unless stated otherwise.

4. Show ALL calculations clearly.

5. Maps and diagrams are NOT drawn to scale, unless otherwise stated.

6. Indicate units of measurement, where applicable.

7. Round off ALL final answers appropriately accordingly to the given context, unless stated otherwise.

8. Start EACH question on a NEW page.

9. Write neatly and legibly.

QUESTION 1

1.1 The table below shows the relationship between hours and the rate of charged per hour or part thereof . Tariffs include value added tax (VAT). Refer to the table and answer the questions that follow.

TABLE 1: PARKING FEES FOR THIS AREA

HOURS	RATE CHARGED PER HOUR OR PART THEREOF:
0–2	R5,00
2–3	R7,00
3–4	R10,00
4–5	R12,00
5–6	R15,00
6–8	R20,00
More than 8 hours	R40,00

NOTE: Lost ticket penalty is R50,00 plus additional charges .

1.1.1 What is the rate charged if Mr Sokutu parked his car for 8 hours 15 minutes ? (2)

1.1.2 Write 8 hours and 15 minutes in hours . (2)

1.2 Mr Titi lost his ticket. When looking at the security cameras, they could see that he arrived at the mall at 11:30 am and that it was now 14:20 pm.

1.2.1 Determine how much time lapsed . (2)

1.2.2 Calculate how much Mr Titi was charged . (2)

1.3 Refer to the rectangular diagram below and answer the questions that follow .

210 cm

Scale 1 : 100

1.3.1 Define the term perimeter . (2)

1.3.2 Determine the perimeter of the rectang ular diagram in centimetres . (2)

1.3.3 Give the name of the scale found on the diagram . (2)

1.3.4 Explain what scale 1 : 100 means . (2)
100 cm

1.4 1.4.1 Identify from the list below a provincial road in South Africa.

They are as follows:

N10
R44
M75 (2)

1.4.2 Define the term provincial road in the above context. (2)
[20]

QUESTION 2

2.1 The tylosaurus proriger was an ancient sea animal that is thought to have lived 85 million years ago. The top speed of the tylosaurus is about 64 km/h. This animal weighed 20 tons.

[Source: https: // kids.national geographic.com]

Refer to the diagram below. Use the scale to calculate the length of the animal below .

1, 9 meters

2.1.1 Write down the name of the scale used above . (2)

2.1.2 Use the scale to calculate the actual length of the animal in metres . (5)

2.1.3 This animal weighed 20 tons. Convert 20 tons to kilograms. (2)

2.2 Due to the global fish crisis only 30 000 ton s of tuna may be caught annually. The
true quantity of tuna that is caught , is closer to 48 000 ton s per year.

Calculate the percentage of illegal fishing in one year. (3)

2.3 Refer to the map below and answer the questions that follow .

2.3.1 Name the metropolitan roads found on the map . (2)

2.3.2 List the major national road that link s Bluewater Bay to Coega . (2)

2.3.2 Identify the provincial road between Ibhayi and Despatch . (2)

2.3.4 How many provincial roads are indicated on the map ? (2)

2.3.5 Mr Thulani stated that the measured distance on the map from Motherwell to Bluewater Bay is 5 cm, and the actual distance is 13 km. He further said that the scale of the map is 1 : 260 000.

Verify , with the necessary calculations , if his statements are valid. (6)

2.3.6 Determine the time (in minutes and seconds) taken by the Thulani family to travel from Motherwell to Bluewater Bay, if they travelled by car at an average speed of 80 km/h for a total distance of 13 km.

Use the formula:

Time = Distance
Average Speed (4)
[30]

QUESTION 3

3.1 The diagram below shows the floor plan of the living room of the Sokutu family 's house.

850 cm

3,5 m
N

Study the information below and answer the questions that follow .

3,5 m

3.1.1 Calculate the perimeter of the living room in mm. Use the formula below:

Perimeter of rectangle = 2 × (length + breadth) (3)

3.1.2 Calculate the area of the floor in m2. Use the formula below:

Area of rectangle = Length × Breadth (3)

3.1.3 If a concrete floor which is 10 cm thick , is to be laid, how many cubic metres of concrete will be needed? You may use the formula:

Volume = Length × Breadth × Height (2)

3.1.4 In which directions does the width of the living room walls face? (2)
Floor plan
Living room dimensions
Length = 850 cm
Breadth = 3,5 m

Height = 10 cm

3.2 This diagram below represents a picture of the southern wall of the Sokutu family's living room. They carefully selected red bricks for the wall. This diagram is not drawn
to scale. The following information indicated on the diagram below is useful for calculation.

[Adapted from vectorstock.com/red -brick wall/]

Southern wall dimensions :
Length = 850 cm
Height = 5 m
Dimensions of ONE brick: height = 73 mm length = 222 mm width = 106 mm
Area of ONE Brick = length × height
NOTE: Mortar is a workable paste which hardens to bind building blocks such as bricks.

3.2.1 Determine the number of bricks needed for the southern wall shown in diagram above.
NOTE : The dimensions of ONE brick exclude the mortar which is 10 mm.

You may use the following formula:

No. of bricks = Area of the wall in m2
Area of ON E brick in m2
(9)

3.2.2 The builder suggested that an additional 10% bricks should be ordered to cover for the wastage. Calculate the total number of bricks needed for the boundary wall. (2)

3.2.3 The price of one brick is R4,75, VAT included. Calculate the total amount to be spend on bricks. (Use your answer in QUESTION 3.2.2.) (2)
[23]

Height = 5 m

QUESTION 4

4.1 Jonah Mouwer buys a small fish tank from Bay Aquariums for the price of R472,00 (VAT inclusive). He wants to present the dimensions of his fish tank on a piece of paper. His father , an architect by profession , told him to use a scale of 1 : 30.

This table below shows the dimensions of a small fish tank. Study the table and answer the questions that follow .

TABLE 2 : MEASUREMENTS OF THE FISH TANK

DIMENSIONS OF THE SMALL FISH TANK IN MET RES	DIMENSIONS ON A PIECE OF PAPER IN CM
Height = 0,45 m	A
Length = 1,05 m	B
Width = 0,3 m	Width = 1 cm

NOTE : Volume = length × width × height

4.1.1 Refer to the table above. Use the scale to calculate the missing value A and B. (5)

4.1.2 Determine the volume of the small fish tank in cm3. (3)

4.1.3 How many litres of water will the tank hold when it is 90% full?

You may use: 1 cm3 = 1 m■ (3)

4.2 The winner was selected randomly. Mr Sokutu won the grand prize of R150 000. He submitted 1 000 entries out of a total of 10 000 entries for the competition. He further decided to use R90 000 from the R150 000 to extend his single garage into a double garage. He had a draughtsman draw up the plan. The plan is drawn below .

Scale 1 : 150

NOTE : Refer to the diagram above and use the scale .

4.2.1 Calculate the actual length and breadth of the existing and planned new garage in metres using the given scale. (6)

Given : Measure length = 50,4 mm
Measure width = 25,6 mm

4.2.2 Calculate the floor space he will now have available in his new double garage. Give your answer to the nearest square metres.
You may use the formula:

Area of Rectangle = Length × Breadth (3)

4.3 Determine the probability of Mr Sokutu not winning the competition , taking into account the number of entries he submitted. Give your answer in percentage format. (3)

4.3.1 What is the probability of Mr Sokutu winning the competition? Choose from the table below.

Impossible	Unlikely	Even Chance	Likely	Certain

(2)

4.3.2 Write 10% to 90% in simplified ratio format. (2)
[27]

TOTAL: 100

NATIONAL
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GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P2
ADDENDUM
(EXEMPLAR)

This addendum consists of 5 pages with a 4-page annexure.

IMLIT4

Copyright reserved Please turn over ANNEXURE A

QUESTION 2.2

ASSEMBLE A – FLOOR LAMP

IMPORTANT: Always disconnect the power before installing or replacing a bulb and before cleaning or other maintenance.

Assembly and installation
instructions

1. Attach the base to stem 3, inserting the threaded stem and locator pin through the washer and in the base. Secure the lock washer and hex nut using the wrench (included).
2. Insert the connector on stem 2 into stem 3 and use the large Allen key to tighten the set screws. (Loosen the set screws to adjust the height of the lamp)
3. Insert stem 1 into stem 2 and use the small Allen key to tighten the set screw.
4. Attach the diffuser, shade, spacer, and harp to the socket, and secure with socket ring.
5. Install a 100- Watt medium base bulb (not included).
6. Attach the glass and flat plate to the harp and secure with the finial.
7. Plug in the cord and the dimmer switch is located on the cord. Floor Lamp

[Source: https://manualzz.com/doc/19886634/assembly-instructions-for-item]

Copyright reserved Please turn over ANNEXURE B

QUESTION 3.1

QUESTION 3.2

RECIPE FOR BEEF STROGANOFF
For the Beef Stroganoff:
x 1 lb (pound) top sirloin steak thinly sliced into strips
x 2 (tbsps.) tablespoon olive oil
x 2 (tbsps.) tablespoon butter
x ■
■ medium onion, finely chopped
x ■
■ lb (pound) brown mushrooms, thickly sliced
x 1 garlic clove minced
x 1 (tbsp.) tablespoon all-purpose flour
x 1 cup beef broth
x ■
■ cup heavy whipping cream
x ■
■ cup sour cream
x 1 (tbsp.) tablespoon Worcestershire sauce
x ■
■ (tsp.) teaspoon Dijon mustard
x ■
■ (tsp.) teaspoon salt
x ■
■ (tsp.) teaspoon black pepper

Instructions

1. Place a large deep pan over medium-high heat. Add 2 tbsps. oil and once the oil is very hot, add thinly sliced beef strips in a single layer, cooking 3 minutes per side without stirring. Cook until just browned and no longer red. Remove beef to a plate and cover to keep warm.

2. Add 2 tbsps. butter, chopped onion and sliced mushrooms. Sauté 6 to 8 minutes or until liquid has evaporated and onions and mushrooms are soft and lightly browned.

3. Add 1 minced garlic clove and sauté 1 minute until fragrant. Add 1 tbsp. flour and sauté another minute stirring constantly.

4. Pour in 1 cup of beef broth, scraping any bits from the bottom of the pan. Then add ■
■ cup of whipping cream and simmer another 1 to 2 minutes or until slightly thickened.
5. Stir a few tablespoons of the sauce into ■
■ cup of sour cream to temper it so the sour cream does not curdle. Then add it to the pan while stirring constantly.

6. Stir in 1 tbsp. Worcestershire, ■
■ tsp Dijon mustard, and season with ■■ tsp salt and ■■ tsp pepper, or season to taste and continue simmering for 20 minutes until sauce is creamy. Add beef with any accumulated juices back to the pan and bring just to a simmer for 2 minutes until beef is heated through.

QUESTION 4.1

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NOVEMBER 2020

MATHEMATICAL LITERACY P2
ADDENDUM
(EXEMPLAR)

This addendum consists of 5 pages with a 4-page annexure .

ANNEXURE A

QUESTION 2.2

ASSEMBLE A – FLOOR LAMP

IMPORTANT: Always disconnect the power before installing or replacing a bulb and before cleaning or other maintenance.

Assembly and installation instructions

1. Attach the base to stem 3, inserting the threaded stem and locator pin through the washer and in the base. Secure the lock washer and hex nut using the wrench (included).
2. Insert the connector on stem 2 into stem 3 and use the large Allen key to tighten the set screws. (Loosen the set screws to adjust the height of the lamp)
3. Insert stem 1 into stem 2 and use the small Allen key to tighten the set screw.
4. Attach the diffuser, shade, spacer, and harp to the socket, and secure with socket ring .

5. Install a 100 -Watt medium
base bulb (not included).
6. Attach the glass and flat
plate to the harp and
secure with the finial.
7. Plug in the cord and the
dimmer switch is located
on the cord. Floor Lamp

[Source: https://manualzz.com/doc/1 9886634/assembly -instructions -for-item]

ANNEXURE B

QUESTION 3.1

ANNEXURE C

QUESTION 3. 2

RECIPE FOR BEEF STROGANOFF
For the Beef Stroganoff:
■ 1 lb (pound) top sirloin steak thinly sliced into strips
■ 2 (tbsps.) tablespoon olive oil
■ 2 (tbsps.) tablespoon butter
■ 1
2 medium onion , finely chopped
■ 1
2 lb (pound) brown mushrooms , thickly sliced
■ 1 garlic clove minced
■ 1 (tbsp.) tablespoon all-purpose flour
■ 1 cup beef broth
■ 3
4 cup heavy whipping cream
■ 1
4 cup sour cream
■ 1 (tbsp.) tablespoon Worcestershire sauce
■ 1
2 (tsp.) teaspoon Dijon mustard
■ 1
2 (tsp.) teaspoon salt
■ 1
4 (tsp.) teaspoon black pepper

Instructions

1. Place a large deep pan over medium -high heat. Add 2 tbsps. oil and once the oil is very hot, add
thinly sliced beef strips in a single layer, cooking 3 minute s per side without stirring. Cook until
just browned and no longer red. Remove beef to a plate and cover to keep warm.

2. Add 2 tbsps. butter, chopped onion and sliced mushroom s. Saut é 6 to 8 minutes or until liquid has
evaporated and onions and mushrooms are soft and lightly browned.

3. Add 1 minced garlic clove and sauté 1 minute until fragrant. Add 1 tbsp. flour and sauté another
minute stirring constantly.

4. Pour in 1 cup of beef broth, scraping any bits from the bottom of the pan . Then add 3
4 cup of
whipping cream and simmer another 1 to 2 minutes or until slightly thickened.
5. Stir a few tablespoons of the sauce into 1
4 cup of sour cream to temper it so the sour cream does
not curdle . Then add it to the pan while stirring constantly.

6. Stir in 1 tbsp. Worcestershire, 1
2 tsp Dijon mustard, and season with 1
2 tsp salt and 1
4 tsp pepper, or
season to taste and continue simmering for 20 minutes until sauce is creamy. Add beef with any
accumulated juices back to the pan and bring just to a simmer for 2 minutes until beef is heated
through.

ANNEXURE D

QUESTION 4.1

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GRADE 11

NOVEMBER 2018

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages.

Imlit1

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FIVE questions. Answer ALL the questions.

2. Number the answers correctly according to the numbering system used in this question paper.

3. An approved calculator (non-programmable and non-graphical) may be used, unless stated otherwise.

4. Maps and diagrams have not been drawn to scale, unless stated otherwise.

5. Round off ALL the final answers according to the context used, unless stated otherwise.

6. Indicate units of measurement, where applicable.

7. Start EACH question on a NEW page.

8. Show ALL calculations clearly.

9. Write neatly and legibly.

QUESTION 1

1.1 A house in East London was for sale at a price of R2 578 799,00. A deposit of R386 819 , 85 was required and the balance payable in equal monthly instalments for 20 years.

1.1.1 Write the sale price of the house in words. (2)

1.1.2 Express the deposit amount as a percentage of the sale price. (3)

1.1.3 Ben decided to bank the deposit amount at QR BANK . Use the information below to calculate the transaction cost of the deposit.

Transaction Rate
Charge for deposits R5,75 + R1,10 per R100 or part thereof (3)

1.2 TABLE 1 below shows Babu's running time during a 2015 Comrade Marathon at various points along the route.

TABLE 1
Athlete: Babu

Points on the route	Distance in kilometres	Time (hours, minutes, seconds)
Lion Park	15,9	01:05:26
Camperdown	26,9	01:50:39
Halfway	45	03:05:14
Pinetown	68,9	04:54:45
Mayville	82,3	06:02:45
Finish	89,3	06:37:30

Use the information in TABLE 1 to answer the questions that follow:

1.2.1 Determine the distance from Camperdown to Mayville. (3)

1.2.2 Calculate how long it took Babu to run from the halfway mark to Pinetown. (2)

1.2.3 Convert the distance to Pinetown to metres. (2)

1.3 The graph below shows the number of houses sold in different locations A, B , C, D and E.

Use the graph above to answer the following questions.

1.3.1 Arrange the number of houses sold in descending order. (2)

1.3.2 Write down the type of graph that was used to represent the information above. (2)

1.3.3 Calculate the total number of houses that was sold in all the locations. (2)
[21]

QUESTION 2

2.1 Ms Fasi started a small business by selling pies at schools and factories close to her workplace. She pays rent of R1 500 per month. It costs her R5,00 to make and package a pie and she sells them at R15,00 each.

Study the table below and answer the questions that follow.

TABLE 2: THE COST AND INCOME FOR THE BUSINESS

Number of Pies	0	50	100	150	250	350
Total Cost in Rand	1 500	A	2 000	2 250	2 750	3 250
Income in Rand	0	750	1 500	2 250	B	5 250

The following formulae are used to calculate the cost and income respectively:

Cost = R1 500 + R5,00 ■■n
Income = R15,00 ■■ n; where n represents the number of pies

2.1.1 Use the table to determine Ms Fasi's break-even amount. (2)

2.1.2 Calculate the value of A. (3)

2.1.3 Show by means of calculations that the value of B is R3 750. (2)

2.1.4 Calculate the profit if 350 pies are sold. (3)

2.2 Ms Fasi borrowed R60 000 from Women's Bank to start her small business and agreed to pay back the money at an interest rate of 8,5% that is compounded annually for 2 years.

2.2.1 Calculate the amount of the interest that was added at the end of the first year.
(2)

2.2.2 Determine the total amount that Ms Fasi paid back to the bank after 2 years. (5)

2.3 TABLE 3 shows the Domestic Water Tariffs for 2018 used by the local municipality where Ms Fasi lives.
TABLE 4 shows the meter readings by the local municipality which indicates the amount of water the Fasi family used for May and June 2018.

TABLE 3

	Number of kilolitres	Cost per kilolitre (kκ) excluding VAT
1	0 – 6 kκ	0
2	Above 6 k κ – less than 30 k κ	R10,02
3	30 kκ – less than 60 k κ	R12,28
4	60 kκ and above	R16,70

+ Additional charge if more than 6 k κ used R80,70

TABLE 4: Water meter readings for Account Number 40101607 during May and June.

1/05/2018 (kκ) 0561
1/06/2018 (kκ) 0587

2.3.1 Calculate the cost of the water usage for May 2018 excluding VAT. (5)

2.3.2 Calculate the VAT amount that is charged on the additional charge of R80,70. (VALUE ADDED TAX = 15%)
(3)

2.4 The average inflation rates for the period 2016 to 2017 are shown in the following table.

	2016	2017
Average inflation rate	4,51%	6,59%
Cost of a brown bread	R9,99	A

2.4.1 Explain the meaning of the term inflation rate . (2)

2.4.2 Calculate the cost of a loaf of brown bread in 2017 by using the average inflation rates that are given in the table above. (2)
[29]

QUESTION 3

3.1

Study the weather forecast for Cape Town and Pretoria on the 20th of May 2018.

Forecast	Cape Town	Pretoria
Sunrise	07:35	06:40
Sunset	17:50	17:27
Humidity (%)	68	58
Visibility (miles)	6,0	12
Maximum Temperature (ιC)	20	17
Precipitation	0	0

Visibility is a measure of the distance at which an object or light can be clearly seen.

3.1.1 Determine the visibility distance in kilometres for Pretoria.
(Use 1,609 km = 1 mile)
(2)

3.1.2 Write down Cape Town 's humidity as a simplified common fraction. (2)

3.1.3 Express the sunset time in Cape Town in 12-hour format. (2)

3.1.4 Convert the maximum temperature for Pretoria to degrees Fahrenheit (°F). Give your final answer to the nearest whole number.

You may use the following formula:
■■=■(■■ ×■1,8) + 32 (3)

3.2 Study the fish tank below and answer the questions that follow.

[Source: quora.com]

3.2.1 Calculate the volume of the fish tank above in cubic inches (in3).

You may use the formula:
Volume = Length × Width × Height

(3)

3.2.2 The fish tank is 85% full. After adding stones to the bottom of the fish tank, the fish tank is 97% full. Calculate the volume of the stones.
(5)
[17]

QUESTION 4

4.1 Rochester is in the north east of New York state. Study the map below and answer the questions that follow.

[Source : filmrochester.org]

4.1.1 Write down the name of the city where Roads 84, 91 and 95 meet. (2)

4.1.2 Using the given scale, determine the actual distance in miles between Buffalo and Albany, if the distance on the map between these two places is 7,5 cm.
(3)

4.1.3 Identify the roads that Bande will use to travel from Scranton to Albany. (2)

4.1.4 Write down the compass direction when travelling from Hartford to Boston. (2)

4.1.5 Identify the road that will help you to travel from New York to the far west of the map. (2)

4.1.6 Determine the probability of randomly selecting a road on the map with an even number. (2)
[13]

QUESTION 5

Study the information on eggs in Incubators and Poults/Chickens hatched during the different months in the United States for the 2016 –2017 period.

Month	Eggs in incubators on the first of the month	Poults/Chickens hatched during the entire month
	2016–2017	2016–2017
September	28 927	23 645
October	28 409	23 572
November	27 179	22 782
December	28 795	25 422
January	29 961	25 332
February	29 906	23 598
March	30 030	25 719

April	28 597	23 179
May	28 825	24 067
June	29 441	25 075
July	29 271	24 616
August	29 725	24 786
TOTAL	--------	---------

5.1 Determine the number of eggs that did not hatch in December. (2)

5.2 Calculate the mean number of eggs in incubators for the 2016 –2017 period. (3)

5.3 Write down the modal value for the number of eggs in the incubators on the first of the month.
(2)

5.4 Determine the range of the Poults hatched the entire month during the 2016 –2017 period.
(2)

5.5 Calculate the median of the Poults hatched for the entire month during the 2016 –2017 period.
(3)

5.6 Express the number of hatched poults during the entire month of March as a ratio to the total hatched during 2016 –2017 period.
(3)

5.7 Determine the probability, as a percentage, of randomly selecting the number of eggs in an incubator during July.
(3)

5.8 Name the type of graph that can best display the above information. (2)
[20]

TOTAL: 100

■
■■
■■
■
■■
■
■■
■
■■
■

NATIONAL
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GRADE 11

NOVEMBER 2018

MATHEMATICAL LITERACY P2
ADDENDUM

This addendum consists of 4 pages with a 3-page of annexures
IMLIT4

ANNEXURE A

QUESTION 2.2

ANNEXURE B

QUESTION 3

ANNEXURE C

QUESTION 4.2

NATIONAL
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GRADE 11

NOVEMBER 2018

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages and an addendum of 4 pages .
imlit2

INSTRUCTIONS AND INFORMATION

Read the following instructions care fully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ADDENDUM with ANNEXURES for the following questions:
ANNEXURE A
for QUESTION 2.2
ANNEXURE B for QUESTION 3
ANNEXURE C for QUESTION 4.2

ANSWER SHEET 1 for QUESTION 2.2.3

Write your name in the space provided on the ANSWER SHEET and hand in the ANSWER SHEET with your ANSWER BOOK.

3. Number the questions correctly according to the numbering system used in this question paper.

4. Start EACH question on a NEW page.

5. An approved calculator (non-programmable and non-graphical) may be used, unless stated otherwise.

6. Show ALL calculations clearly.

7. Round off ALL final answers appropriately accordingly to the given context, unless stated otherwise.

8. Indicate units of measurement, where applicable.

9. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

QUESTION 1

1.1 Mrs Day has a laundry business in town and charges:
x R16 per kg washing and drying
x R3 for Staysoft (fabric softener) per 5 kg
x R25 per kg for hand wash with Staysoft (fabric softener)
x R20 per 5 kg for ironing

1.1.1 How much will a person with 18 kg of washing pay for washing and drying with Staysoft (fabric softener) and ironing? (5)

1.1.2 A family has a helper coming on Mondays and Fridays for the month shown below for washing. They pay the helper R150 per day for washing and drying without ironing.
Use the calendar for March 2018 to answer the questions.

(a) The family claims that if they can take their washing every Friday to the laundry for washing and drying with Staysoft (fabric softener), instead of hiring a helper , they could save R70 per month. Verify, showing all calculations , whether their claim is valid if they have an average of 15 kg of washing and drying and 2 kg of hand wash for Fridays in March 2018. (8)

(b) Calculate the probability of having a Friday in March with a date as an even number. Give your final answer as a percentage to one decimal place. (3)

1.2 To cater for hand washing and to save water, the laundry uses water collected from rain and stored in cylindrical storage containers, as shown in the figure below. They have 10 such containers.

1.2.1 Show, with calculations, whether the 10 containers can fit on the floor of a storeroom which is 3 m by 2,5 m. (5)

1.2.2 For hand washing, they use 100 litres of water per 25 kg of washing. The average of hand washing is 30 kg per day.

You may use formula:

Volume of cylindrical container =
π × radius × radius × height ; where π = 3,142
Given that 1 litre = 1 000 cm3

(a) Calculate the capacity of the container as shown in the diagram. (3)

(b) The supervisor claims that five containers of water will be enough for 5 days. Verify with calculations, whether the claim is valid or not. (5)

1.2.3 Provide ONE possible reason why some clothes are hand washed. (2)
[31]

70 cm
50 cm

QUESTION 2

2.1 A printing company has employed people for typing and editing material to be used in workshops. They are paying the editors R97 to edit one document.
For typists, they use the following formula :
Amount paid for typist = ■■■■■■■■■■
■■ × rate × number of documents typed

NOTE : Norm time is the time in minutes spent on typing and is 28 minutes per document. The rate is R195 per document.

The employer is claiming that he is paying an editor 5% more than a typist for every 100 documents edited. Verify, with calculations, whether the statement is valid or not. (8)

2.2 Use ANNEXURE A that shows the information on the amount of sugar in different types of drinks.

2.2.1 Determine the range of the number of grams of sugar for all the 330 m κ drinks. (3)

2.2.2 In a hotel, guests are allowed two drinks per person. One guest has a Play energy drink and a Powerade. Another guest has an Appletiser and a Coca-Cola.

The guest drinking the energy drinks , claims she is taking 2 teaspoons of sugar less than the other guest. Verify, with calculations, whether the statement is valid or not. (6)

2.2.3 Use the ANSWER SHEET provided, and draw a bar graph showing the amount in grams for the following drinks only:

(a) Bonaqua flavoured water
(b) Glaceau Vitamin water
(c) Sparletta Iron Brew
(d) Fanta Orange
(e) Coca-Cola (5)
[22]

QUESTION 3

3.1 Use the distance table in ANNEXURE B to answer the following questions.

3.1.1 A couple is travelling from Durban to Nelspruit at a speed of 110 km per hour. How long will this journey take them? Give your final answer to the nearest hour and minute.

You may use the following formula:
■■■■■ ■■■■■■■■■
■■■■ (6)

3.1.2 Determine the median distance between Pretoria and all the other towns and cities. (4)

3.1.3 Calculate the difference between the mea n distance for all towns to Pretoria and the mean distances for all towns to Port Elizabeth. (7)

3.1.4 To travel by bus, cost 70c per km. Ca lculate how much a person will pay for transport from Bloemfontein to Cape To wn. Give your final answer rounded off to the nearest rand. (3)
[20]

QUESTION 4

4.1 A netball court with dimensions of the centre circle and semi-circles at the extreme ends are given in the diagram below.

The netball court has the following dimensions:
Length of the court = 30,5 m
Breadth of the court = 15,25 m
Diameter of centre circle = 0,9 m
Radius of semi-circle = 4,9 m

Calculate the difference between the area of the centre circle and the area of one of the semi -circles at the extreme ends of the netball court.

You may use the following formulae:
Area of circle = ■■■■■■■■■■■■■■■
Area of semi-circle = ■■■■■■■■■■■■■■■■■■■
■; where ■ = 3,142 (5)

4.2 ANNEXURE C shows the seating plan of a sc hool hall. Use the seating plan to answer the following questions.

4.2.1 After the match, the school has an entertainment activity for the guests in the school hall. How many seats are on the left side of the stage? (3)

4.2.2 Calculate the probability of having a row on the right side of the stage with 15 seats. (2)

4.2.3 The hall is sometimes used for hiring. The cost for seating in the middle of the hall is 8% more than the seats on the left and right side of the hall. For a certain function, the tickets are R150 f or a seat on the sides of the hall.

The board claims that if all seats are occ upied, they will be able to collect more than R100 000. Verify, with calculations, whether the statem ent is valid or not. (8)

4.3 TABLE 1 below shows the number of learners at Komga High School in Grade 10, 11 and 12.

TABLE 1

Grade	Boys	Girls	Total
10	73	85	158
11	A	73	137
12	54	45	99
Total	191	203	B

4.3.1 Calculate the number of boys in Grade 11 as a percentage of the total number of learners. (5)

4.3.2 Describe the trend and give a possi ble reason for the trend observed in the total number of learners from Grade 10 to Grade 12. (4)

[27]

TOTAL: 100

ANSWER SHEET for QUESTION 2.2.3

NAME and SURNAME: ……………………………………………………………………….

GRADE 11: …..

051015202530354045
Bonaqua Flavoured
WaterGlaceau Vitamin
WaterSparletta Iron Brew Fanta Orange Coca-ColaAmount of sugar in grams
Different DrinksAmount of sugar in grams for different drinks

■
■■
■■
■
■■
■
■
■■
■
■

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2017

MATHEMATICAL LITERACY P2
ADDENDUM

MARKS: 100

TIME: 2 hours

This addendum consists of 4 pages with 3 annexures .
IMLIT4

ANNEXURE A for QUESTION 4.1

Population group	Male Number	% of male population	Female Number	% of female population	Total Number	% of total population
African	21 168 700	80,3	22 165 000	B	43 333 700	80,2
Coloured	2 305 800	8,7	2 465 700	8,9	4 771 500	8,8
Indian/ Asian	677 000	2,6	664 900	2,4	1 341 900	2,5
White	2 214 400	8,4	2 340 400	8,5	4 554 800	8,4
Total	A	100,0	27 636 000	100,0	54 001 900	100,0

ANNEXURE B for QUESTION 4.2

ANNEXURE C for QUESTION 4.3

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2017

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 7 pages including an addendum of 4 pages .

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ADDENDUM with ANNEXURES for the following questions:

ANNEXURE A for QUESTION 4.1
ANNEXURE B for QUESTION 4.2
ANNEXURE C for QUESTION 4.3

3. Number the questions correctly according to the numbering system used in this question paper.

4. Start EACH question on a NEW page.

5. An approved calculator (non -programmable and non -graphical) may be used, unless stated otherwise.

6. Show ALL calculations clearly.

7. Round off ALL final answers appropriately accordingly to the given context, unless stated otherwise.

8. Indicate units of measurement, where applicable.

9. Maps and diagrams are NOT drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

QUESTION 1

Sipho has a car washing business in the Ziphunzana area. They charge R30 per car and R40 per microbus. They wash on average 10 cars and 5 microbuses per day from Monday to Friday and on Saturdays and Sundays they wash 60% more cars and 30% more microbuses.

1.1 If they wash both cars and microbuses from Monday to Friday, calculate the probability that they will wash a car . (3)

1.2 Sipho claims that if they wash both cars and microbuses from Monday to Sunday , they will be able to receive more than R4 000. With calculations, show whether his claim is valid or not. (10)

1.3 Sipho and his employees use three 25-litres containers of water for a car and four 25-litres containers of water for a microbus. How many litres of water do they use in 7 days? (6)

1.4 Use the table below to answer the question.

TABLE 1: Water tariff charges per month for Ziphunzana in 2014 –2015

Tariff Summary	Tariff R/kl 2014/15 (without VAT)
0 – 6 k■	R8,66
7 – 15 k■	R10,02
16 – 30 k■	R12,28
31 – 45 k■	R15,25
46 – 60 k■	R16,70

NOTE :
1 000 litres = 1 kilolitre
VAT = 14%

Calculate how much Sipho will pay for water per month including VAT.

NOTE : Assume there are 4 weeks in a month . (8)

1.5 Give a possible reason why they wash more cars on Saturdays and Sundays. (2)
[29]

QUESTION 2

2.1 Use TABLE 2 below showing different options with premiums paid and benefits of an insurance company.

TABLE 2

Options with premiums to be paid and benefits of an insurance company

	Option 1		Option 2		Option 3		Option 4	
	Benefit	Premium	Benefit	Premium	Benefit	Premium	Benefit	Premium
Main member	R10 000	R45	R15 000	R65	R20 000	R82	R30 000	R110
Main member and children	R10 000	R63	R15 000	R91	R20 000	R113	R30 000	R140
Main member and spouse	R10 000	R76	R15 000	R107	R20 000	R135	R30 000	R188
Main member, spouse and children	R10 000	91	R15 000	R125	R20 000	R163	R30 000	R212

NOTE:

- Benefits in the table represent money paid out in the event of death for a person 15 years and older
- Benefits for those younger than 15 years are 75% of the amounts indicated in the table.
- Premium is the amount paid monthly to be able to get the death benefit.

2.1.1 What will be the premium paid by a member with a spouse and children to get the benefit of R30 000? (2)

2.1.2 Mary claims that the percentage increase for the premium paid in Option 1 to Option 4 for a member and spouse is more than 100%. Verify, with the necessary calculations, whether Mary’s statement is valid or not. (4)

2.1.3 Give a reason why less money is paid out for a 5 -year-old than for a 15 -year-old. (2)

2.2 Mr May has cut glass into shapes from a circular piece of glass as shown below .

\

Diameter of the circular piece of glass = 150 cm

2.2.1 Write the distance of B to the distance of A as a ratio. (6)

2.2.2 Glass is sold at R15 per square metre excluding VAT. With the necessary calculations, show that the cost of unused glass (including VAT) from the circular piece of glass to cut the cross shaped piece of glass, is less than R20.

You may use the followin g formulae:

- Area of cross shaped part = Area of bigger rectangle + 2 × Area of smaller rectangles
- Area of a rectangle = length × width

■ Area of a circle = π × r2; where π = 3,142 (10)
[24]

C
C

C
C
C
C

B
B
A
30
cm
30 cm
30 cm
30 cm
100 cm
100 cm
C

QUESTION 3

A compound bar graph below represents salaries of six employees in R100’s for two companies. Use the graph to answer the questions below.

EMPLOYEE

3.1 Moesha claims that the mean salary of Company A , is R1 000 more than the mean salary of Company B. Verify, with calculations , whether her statement is valid or not.
(6)

3.2 What is the modal salary of Company B? (2)

3.3 Thato, a worker at Company A receives the modal salary. He lends 15% of one month’s salary to a friend to be paid back over two years . The friend must pay back the money with interest at an interest rate of 5% compounded annually. How much will he receive after two years? (6)

3.4 What is the probability of randomly choosing an employee who earns a salary of less than R8 000 at Company A? (2)
[16]

Salaries in R100’s

QUESTION 4

4.1 The table in ANNEXURE A shows the mid -year population estimates for South Africa by population group and gender in 2014. Answer the questions below that are based on the table in ANNEXURE A.

4.1.1 Calculate the missing values A and B respectively in the table. (4)

4.1.2 Which population group has fewer females than males? (2)

4.1.3 Write the total number of white males in words. (2)

4.2 A seating plan of the Canyon Theatre is given in ANNEXURE B. Answer the questions below based on the seating plan.

4.2.1 What is the difference in the number of seats between the left -hand side and the right -hand side of the theatre? (4)

4.2.2 Determine the seats for the handicapped as a percentage of the total number of seats on the right -hand side of the theatre . Your answer should be rounded off to two decimal places. (3)

4.3 An extract of a map of the WILD COAST is given in ANNEXURE C. Answer the following questions that are based on the map.

4.3.1 Anda travels on the N2 route between Butterworth and Mount Frere. She claims that if she travels at an average speed of 105 kilometres per hour and leaves Butterworth at 7:00 am, she will be able to be on time in Mount Frere for a meeting that starts at 9:30 am. With the necessa ry calculations, show whether her claim is valid or not. Give your answer in hours and minutes.

You may use the following formula:

Speed = ■■■■■■■■■
■■■■ (8)

4.3.2 Anda works for a company which pays her a rate of R2,82 per kilometre for transport cost. This is a rate after an increment of 6,8%. How much did she get for a return trip between Butterworth and Mount Frere before the increment? Express your answer to the nearest rand. (6)

4.3.3 Which other roads on the map except for the N2 can be used to travel on? (2)
[31]

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOV EMBER 201 6

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 10 pages including 3 annexure s.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Use the ADDENDUM with ANNEXURES for the following questions:

ANNEXURE A for QUESTION 2.2

ANNEXURE B for QUESTION 3.1.1 and
ANNEXURE C for QUESTION 3.1 .2

3. Number the questions according to the numbering system used in this question paper.

4. An approved calculator (non-programmable and non -graphical) may be used, unless stated otherwise.

5. ALL calculations must be shown clearly.

6. Round off ALL final answers appropriately accordingly to the given context , unless stated otherwise.

7. Start EACH question on a NEW page.

8. Write neatly and legibly.

QUESTION 1

1.1 Anna has a small business for baking and selling muffins.
She has the following recipe that makes 12 muffins as well as the costs for making the muffins:

500 g muffin mix at R12, 00 per kg
1 large egg at R10, 50 for ½ dozen
½ cup oil (125 m■) at R12, 50 per 750 m■
1 ½ cups of milk (375 m■) at R11, 50 per litre

Preheat oven to 180 °C then bake for 25 minutes.

Anna uses pans that bake 12 muffins and an oven that allows two pans at a time. She sells her muffins at R30 per dozen .

1.1.1 Mention ONE other expense in addition to cost of ingredients that Anna will have when making these muffins. (2)

1.1.2 How much will she spend on ingredients for 24 muffins? (5)

1.1.3 Anna tells a friend that if she would get an order for 1 200 muffins, she would make a profit of more than R1 000. Verify using calculations to prove whether her statement is true or false. (Costs considered are those for the ingredients only .) (5)

1.1.4 Anna has an order for 240 muffins which need to be baked and delivered 50 km from her home where she bakes. She starts baking at 9:00, allows 10 minutes for cleaning pans after each batch of muffins. If she leaves after the last cleaning and drives at 100 kilometres per hour , will she be able to deliver the muffins for tea to be served at 16:00?

You m ay use the formula: Time = Distance
Speed (6)

1.2 After being in business for a year Anna manage d to get a small business loan from the government to expand her business. She ha d to employ staff to help her, and she manages to employ people for 5 hours a day as follows :

- A cleaner at R50 per hour
- A driver at 75% more than the cleaner per hour
- An assistant (baking) at R25 more than the cleaner
- They work 20 days per month

Calculate the total salaries of all the workers for one month . (5)
[23]

QUESTION 2

2.1 A Mathematical Literacy teacher has the following test results for a test out of 50. His class has 14 learners and the results are as follows :

- ■ 4 learners scored 42 which was the maximum mark obtained by the learners
- ■ 3 learners scored 7 marks less than the maximum mark
- ■ 2 learners scored 34 marks,
- ■ 1 learner scored 20 marks less than the maximum mark
- ■ 2 learners scored 16 marks
- ■ 1 learner scored 5 marks
- ■ The last learner scored ½ of the maximum mark

2.1.1 If the percentage pass for the test was 30%, what can you conclude about the performance of the learners in the test? Use calculations on which you base your conclusion. (4)

2.1.2 What is the mean score of the marks? (3)

2.1.3 Calculate the numerical value of the difference between the median and the mode . (4)

2.1.4 What is the probability of a learner scoring below 50% in this test? Give your answer as a percentage. (3)

2.2 In a school there are equal number of Mathematics and Mathematical Literacy students and they are all seated in the hall as shown in ANNEXURE A.

- ■ The Mathematics students are seated in the odd numbered rows and the Mathematical Literacy students are in the even numbered rows.
- ■ Students are seated one behind the other and their student numbers are in ascending order from ML ■ 1 for the Mathematical Literacy students and MAT ■ 1 for the Mathematics students.
- ■ When a row has been filled i.e. row A1 – I1, the students starts to fill the next row starting at A2 to I2 and then fill the next row starting at A3 and so on .
- ■ Occupation of seats starts from row one .

2.2.1 How many Mathematical Literacy and Mathematics students are seated in the middle section of the hall? Use calculations on which you base your conclusion. (3)

2.2.2 Calculate the mean score that was achieved. (3)

2.2.3 Calculate the difference in the number of seats between the seats on the left and the right sections of the hall . (3)

2.2.4 If the Department of Education requires 1 invigilator for every 30 learners, how many invigilators must be appointed? (4)
[27]

QUESTION 3

3.1 Two matric learners managed to get the following set of scores for their Grade 12 examinations :

Subject	Learner A %	Learner B %
IsiXhosa Home Language	75%	67%
English First Additional Language	63%	68%
Mathematics	54%	64%
Life Orientation	72%	76%
Accounting	54%	48%
Economics	45%	60%
Business Studies	74%	69%

3.1.1 By using the information on ANNEXURE B, calculate the APS scores for both learners to show which learner has the highest score. (6)

3.1.2 Draw a compound bar graph on ANNEXURE C that represents the scores of the two learners in IsiXhosa Home Language, Mathematics, Accounting, Economics and Business Studies. (5)

3.1.3 At a certain university the APS score is the only consideration for admission to a certain faculty and they require an APS score of 30. Which of the two learners has a greater chance of being accepted by that university and faculty? (2)

3.2 Mr and Mrs Jones's daughter, a first-year student did not manage to get accommodation on campus and gets private accommodation.
The costs are:
- Single room is R2 200 per month
- Double room is R1 650 per month
- Rent is payable for 11 months

She has to cook for herself and her parents send R1 500 per month for food for ten months.

3.2.1 Calculate the total amount spent for the year on rent and food if she chooses a double room. (4)

3.2.2 Staying in a double room costs less than staying in a single room. Give ONE reason for the difference in costs. (2)
[19]

QUESTION 4

4.1 Learners have a project of painting one side of their block of classes. The block has four classrooms with windows designed as shown below. They only need to paint the sides with these windows and there is only one such side in each classroom. Drawing not drawn to scale.

- Dimensions of the classroom are: length = 1 200 cm and height = 500 cm
- Height of the triangular window is 100 cm and the base is 150 cm.
- Width of the square window is equal to the base of the triangular window.
- Radius of the circular window is ½ (half) the base of the triangular window.

1 200 cm

You may use the following formulae:
■ Area of rectangle = length x breadth
■ Area of square = side2
■ Area of triangle = ½ base x height
■ Area of circle = ■ x radius2

Use ■ = 3,142

Calculate the total area of the walls that are to be painted in all the classrooms to the nearest square metre (m2). (14)

4.2 Paint is sold in 5 litre and 20 litre tins . The cost is R105 for the 5 litre tin and R405 for the 20 litre tin. Paint covers 8 m2 per litre.

Learners argue that it will be cheaper to buy 5 litre tins. Using calculations, prove whether their argument is valid or not . (6)

500 cm

4.3 5 litre paint tins are packed in shops for delivery in big rectangular boxes . An example of these boxes are shown below. The diagrams are not drawn to scale.

24 cm

There are two rectangular boxes and their dimensions are as follows :
Dimensions of Box A
Length = 130 cm
Width = 104 cm
Height = 25 cm
AND
Dimensions of Box B
Length = 65 cm
Width = 52 cm
Height = 49 cm

Dimensions of 5 litre paint tin:
Radius = 9 cm
Height = 24 cm

4.3.1 The packers argue that if they use Box A for packaging the paint tins, they will be able to pac k more than two times the paint tins in Box B. Use calculations to prove if their argument is valid or not . (6)

4.3.2 A truck carrying 20 boxes of Box B with 5 litre paint tins has an accident and all the paint is spilt. How much in Rand will be lost ? (5)
[31]

TOTAL: 100
BOX A
130 cm
25 cm
65 cm
49 cm
9 cm

BOX B

ANNEXURE A: QUESTION 2.2

Seating plan of school hall.

ANNEXURE B: QUESTION 3.1.1

Admission requirements for students obtaining the National Senior Certificate (NSC) or Independent Examinations Board (IEB) in 2013 and thereafter:

APS National

NCS IEB
10
9
8
7 (80■100%) 7
6 (70■79%) 6
5 (60■69%) 5
4 (50■59%) 4
3 (40■49%) 3
2 (30■39%) 2
1 (0■29%) 1

*Life Orientation must be divided by two and rounded up to the nearest 10% to calculate the total APS .

Additional information:

■ APS stands for Admission Point Score.
■ Point s are assigned to the percentages obtained in each of the seven NCS subjects
■ Overall APS is calculated by giving the sum of all the points obtained.
■ Admission to a certain field of study depend on whether the poin ts for that particular field have been achieved.

ANNEXURE C: QUESTION 3.1.2

NAME:

GRADE 11:

05101520253035404550556065707580Percentages per subject
SubjectsPercentages of five subjects for two Grade 12 learners
IsiXhosa Maths Accounting Economics Business Studies

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2015

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 12 pages, including an annexure of 3 pages.

imlit2

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. QUESTION 1.1 must be answered with reference to ANNEXURE A, QUESTION 2.1.2 must be answered on ANNEXURE B and QUESTION 4.1.7 must be answered on ANNEXURE C. Write your NAME and CLASS GROUP on ANNEXURES B and C and hand in with your ANSWER BOOK.

3. Number the questions correctly according to the numbering system used in this question paper.

4. An approved calculator (non-programmable and non-graphical) may be used, unless stated otherwise.

5. ALL calculations must be clearly shown.

6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. Start EACH question on a NEW page.

8. Write neatly and legibly.

QUESTION 1

1.1 Study the floor plan of the Greenacres Shopping Centre in Port Elizabeth (ANNEXURE A) and answer the questions that follow.

1.1.1 How many entrances can you use to access the centre? (2)

1.1.2 You get a call from your friend. The two of you lost each other in the centre. Your friend is at the Automatic Teller Machine (ATM) at Shop No. 40. You are waiting for your friend at Entrance No. 2. Explain to your friend the quickest way how to get to Entrance No.2. (3)

1.2 You and your friend want to watch the movie "The Best of Me" that is screened in Cinema 15 at "The Bridge" near the Greenacres Shopping Centre. The running time of the movie is 117 minutes.

The show times of the movie are shown below for Sunday, 11 January 2015.

Sun 11 Jan					
10:00	12:15	14:45	17:45	20:15	22:30

1.2.1 Write the running time of the movie in hour s and minutes. (2)

1.2.2 You and your friend arrived at the centre at 11:20 and have to be at home by 17:30. It takes you two thirds of an hour from Greenacres Shopping Centre to your home. With your arrival at the centre, you and your friend spend three quarters of an hour at the Food Court. It will take you about 15 minutes to buy your tickets. Show with the necessary calculations which of the times will be most suitable for you to watch the movie. (6)

1.2.3 Give TWO reasons why the movie for the next time slot is not exactly after the end of the previous screening. (2)

1.3 The following table shows the prices of the movie tickets at The Bridge Cinemas:

Kids (younger than 18 years)
Adults (18 –60 years)
Seniors (older than 60 years)

Table 1: Price of movie tickets at The Bridge

Regular Pricing	Adults	Kids and Seniors
2D	R■62,00	R43,00
3D	R■75,00	R55,00
– Mon, Tue, Thurs, Fri		
2D	R■38,00	
3D	R■55,00	
Wowza Wednedays		
2D	R■37,00	
3D	R■45,00	

1.3.1 Calculate the total price of the tickets for you and your friend. You are 18 years old and your friend 16 years; and you will be watching the movie in 3D. (3)

1.3.2 Seeing that you and your friend are dependent on allowances from your parents, why d o you think you and your friend chose the wrong day to watch movies? (2)

[20]

QUESTION 2

2.1 Mr. Gardner has a landline telephone. His service provider offers him a choice of their new call packages. Below is a table showing the two different call packages that they offer.

Table 2: Two Landline call packages

Call Package A	Call Package B
■ Monthly rental of R100 ■ First 100 minutes are free ■ Calls cost 80 cents per minute	■ Monthly rental of R200 ■ First 150 minutes are free ■ Calls cost 60 cents per minute

2.1.1 The total cost for Call package A is given by the following formula:

Total cost (in Rand) = R100 + number of minutes more than 100 x R0,80

(a) Write down a formula which can be used to calculate the total cost in Rand for Call package B. (3)

(b) If Call package B is used, determine the total cost in Rand, if Mr. Gardner and his family made calls with a total duration of 625 minutes. (4)

2.1.2 The line graph illustrated in ANNEXURE B shows the total cost for Call package B. On the same set of axes, draw a line graph to illustrate the total cost for Call package A. (5)

2.1.3 Briefly define what is meant by concept , break -even point , in the given context. (2)

2.1.4 Write down the co -ordinates of the break -even point on the graph. (2)

2.1.5 Show by means of calculation that Call Package A costs 20 cents more than Call Package B for using the minimum number of minutes after the break -even point . (4)

2.1.6 Mr. Gardner wants to spend a maximum of R300 per month on one of the Call packages. Which Call package would you advise him to choose? Motivate your answer by showing all calculations. (5)

2.2 The service providers have a major task on their hands. For Landline numbers they have to use a system where they make entirely sure that a number is not duplicated (repeated).

Landline numbers consists of ten digits. The first three digits are for the a rea code followed by seven digits. The first three digits of the remaining seven digits for the area where Mr. Gardner lives, starts with 423 followed by the last four digits.

The last four digits are numbers that can be repeated using the numbers 0 to 9.

Show with calculations h ow many households in Mr. Gardner's area can be issued with a landline number starting with 041 423 _ _ _ _? (3)
[28]

QUESTION 3

3.1 As availability of water is becoming a major problem in South Africa, more people are intending to save water. Therefore some of them are investigating how they are billed and if the billing is correctly calculated.

Mrs. Treace is one of those people and stays in Cape Town. She studied the water tariffs for 2013 –2014 carefully.

Below is a table showing the water tariffs for Cape Town during 2013 –2014. Study the table carefully before answering the questions.

Table 3 Water Tariff for Cape Town 2013 –2014

From	To	Rand per kl excluding VAT	Including VAT
> 0,0 kl	6,0 kl	R 0,00	R0,00
> 6,0 kl	10,5 kl	R7,60	R8,66
> 10,5 kl	20,0 kl	R11,61	R13,24
> 20,0 kl	35,0 kl	R17,20	A
> 35,0 kl	50,0 kl	R21,24	R24,22
> 50,0 kl		B	R31,95

3.1.1 Calculate the missing values for A and B respectively. (4)

3.1.2 If Mrs. Treace uses on average 19,5 kl of water per month, will she be paying for the full 19,5 kl? Give a reason for your answer. (2)

3.1.3 Hence, calculate how much Mrs. Treace will pay for using 19,5 kl of water including VAT. (4)

3.2 One of the ways to save water is to install a water tank. Mrs. Treace decided to investigate the different types of water tanks that is sold in order to make the best decision.

Below are diagrams of a vertical , round water tank and a rectangular water tank. Study these diagrams and the information given and then answer the questions that follow.

Vertical Round water tank Rectangular water tank

Vertical Round water tank Rectangular water tank
d = diameter
h = height
f = recommended filled height l = length
w = width
h = height
f = recommended filled height

Vertical Round Water Tanks

Tank	Volume in litres	Diameter in mm	Height in mm
1	1 000	750	2230

2 1 000 1100 1400
3 1 000 1100 1300

Rectangular Water Tanks

Tank Volume in litres	Depth in mm	Height in mm	Width in mm
1000	520	1770	1260

3.2.1 Show with the necessary calculations that the volume of the vertical round water tank numbered 1 is not exactly 1000 li ters as indicated in the table.

Use the formula:
Volume of vertical round water tank = 3,142 x radius2 x height of the water tank.
Pleas e Note: 1 m3 = 1000 l (6)

3.2.2 With reference to your answer in QUESTION 3.2.1, why do you think that there is a difference in the volume between what you have calculated and the given volume? Also explain how this given information can be misleading to consumers. (3)

3.2.3 Briefly explain why the diagrams indicate a recommended filled height. (2)

3.2.4 In her backyard, Mrs. Treace has only one open area next to her house to install the water tank. The width of the area is 1,15 m. Which of the water tanks must she buy that will satisfy the requirement? (2)

3.3 Mrs. Treace will be using the water in her water tank for various purposes. If Mrs. Treace wants to illustrate her usage graphically, which graphical representations can Mrs. Treace use to illustrate how she is using the water? (2)
[25]

QUESTION 4

4.1 The following table illustrates the weather forecast for George from Wednesday, 14 January 2015 to Tuesday, 20 January 2015. Study the weather forecast and answer the questions that follow.

Table 4: Weather forecast for George
Wed
26 °C
Partly
Cloudy Thu
29 °C
Sunny Fri
■
22 °C
Windy Sat
■
23 °C
Partly
Cloudy Sun
■
22 °C
Sunny Mon
■
21 °C
Mostly

Cloudy Tue
24 °C
Partly
Cloudy
Wed
21 °C
Mostly
Clear Thu
20 °C
Partly
Cloudy Fri
■
15 °C
30%
Chance
Rain
Shower Sat
■
16 °C
Mostly
Clear Son
■
17 °C
Partly
Cloudy Mon
■
17 °C
Mostl y
Cloudy Tue
17 °C
Mostly
Cloudy
[http://weather.weatherbug.com]

4.1.1 Calculate the mean day temperature for George for the forecasted period. Round your final answer to the nearest temperature in °C. (4)

4.1.2 Determine the range for the period. (2)

4.1.3 Calculate the median for the night temperatures. (2)

4.1.4 Briefly explain the trend of the night temperatures. (2)

4.1.5 On which day was the difference in temperature the smallest? Write down this difference. (2)

4.1.6 Write down the day where the day and night temperatures showed a decreasing trend. (2)

4.1.7 Use the information in the table to draw a compound bar graph in ANNEXURE C. (6)

4.2 Explain what is meant by the forecast for Friday that states “30% chance of
rain shower”. (2)

4.3 4.3.1 You want to travel from George to Knysna.

The distance between George and Knysna is 64,7 km. Show with the necessary calculations how long it will take you from George to Knysna if you are driving at a speed of 90 km per hour. Give you answer to the nearest minute.

You may use the formula:
Distance = Speed x Time (3)

4.3.2 If you experience problems with your car on the route, will it be possible that you will complete the trip in the time that you have calculated in QUESTION 4.3.1. (2)
[27]

TOTAL: 100

ANNEXURE A

ATM (Automatic Teller Machine)
OTM (Outomatiese Teller Masjien)   Entrance
Ingang

ANNEXURE B

QUESTION 2.1.2

NAME:

GRADE 11:

ANNEXURE C

QUESTION 4.1.7

NAME:

GRADE 11:

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2014

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 14 pages .

INSTRUCTIONS AND INFORMATION

1. This paper consists of FIVE questions. Answer ALL the questions. QUESTION 5.1.3 must be answered on the ANSWER SHEET provided.

2. Number your answers correctly according to the numbering system used in the question paper.

3. A non - programmable and non -graphical calculator may be used, unless stated otherwise.

4. ALL calculations and steps must be shown clearly.

5. Units of measurement must be indicated where applicable.

6. Start EACH question on a NEW page.

7. Write neatly and legibly.

8. Round off ALL FINAL answers to the appropriate form of rounding and/or number of decimal places for a given context.

QUESTION 1

1.1 Johnson worked in London for thirteen years and came back during the December holidays in 2013 to South Africa . Johnson wanted to start a garden service business in March 2014. He would employ two men to assist him. On ANNEXURE A find the quotation he obtained from Rio Hardware Store to answer the questions below:

1.1.1 Calculate his total cost, A, excluding VAT that he will pay for the equipment. (3)

1.1.2 The high density (H/D) plastic refuse bags are sold in rolls of 20 bags per roll. Determine the price of ONE plastic refuse bag. (2)

1.1.3 Write down the month in which Johnson requested the quotation. (2)

1.1.4 Write down the VAT number that Johnson has been issued by SARS (South African Revenue Service) . (2)

1.1.5 Determine what the item is that has an SKU number of RBC -5622. (2)

1.1.6 Write down the quotation number Johnson will have to use if he decides to buy the equipment. (2)

1.1.7 The two stroke oil is sold in bottles of 200 m■. Convert this volume to litres. (2)

1.1.8 Determine the number of months that Johnson spent in London. (2)

1.2 Johnson investigated the school fees charged by different primary schools in order to choose the best school for his child.   The graph below indicates the annual fees charged by schools in his area.

1.2.1 Identify the school with the lowest school fees. (2)

1.2.2 Calculate the difference in the amount of fees between School A and School B. (2)

[21]

R 0.00R 2 000.00R 4 000.00R 6 000.00R 8 000.00R 10 000.00R 12 000.00R 14 000.00R 16 000.00

School A School B School C School D School EFEES PAYABLE AT BAY SCHOOLS PER ANNUM

QUESTION 2

2.1 Johnson had to buy a bakkie in order to do his business. He bought his bakkie for R192 600, 00 including Value Added Tax (VAT) at Sonny's Motors. (VAT = 14%)

2.1.1 Calculate the price of this bakkie excluding VAT. (3)

2.1.2 Johnson paid R478, 20 for 36 litres of petrol for his bakkie. Calculate the price of petrol per litre. (2)

2.2 The following is a summary of Johnson's first month's income and expenses:

TABLE 1: Income and Expenses for Johnson's first month

INCOME : R7 200,00 EXPENSES
Salaries : R2 700,00
Petrol :   R   610,20
Oil: R   28,94

TOTAL: R3 339,14

2.2.1 Show by calculations that Johnson's profit of his business is R3 860,86. You may use the following formula:

Profit = Income – Expenses (2)

2.2.2 Calculate how much an assistant who earns R350, 00 a week contributes towards the Unemployment Insurance Fund monthly if 1% is deducted from his salary . (3)

2.2.3 He increased the price of cutting grass per month from R360,00 to R380,00. Calculate the percentage increase. You may use the following formula:

% increase = new price per cut – old price per cut
old price per cut x 100 (3)

2.3 Luntu has a gold credit card which he used to pay for entertainment during the December 2013 holidays. His total debt amounted to R4 529,96. The bank charges him an interest rate of 27% per annum as well as a monthly subscription fee of R21,00.

2.3.1 Calculate the daily interest rate charged by the bank. (3)

2.3.2 Calculate the instalment due on his card for January 2014 by using the following formula:

Instalment due = amo nt o ed x 1
1 + subscription fee (3)

2.3.3 The cost of a flight back from London to Johannesburg was £379,83. The cost for his flight was charged to his credit card and on the statement an amount of R6 879,00 was charged. Determine the exchange rate that was applicable on the date his flight was booked. (3)
[22]

QUESTION 3

3.1 Nadia High School approached The Winch Company to sponsor them with the mat erial to fence and plant grass o n their sports field and also paint the inside walls of a classroom. The inside walls of a classroom are 6,2 m by 5,1 m and 3 m high. The caretaker indicates that he will need the following material for the work:

Fencing poles, fencin g wire, cement and stone, paint and grass seed.

3.1.1 To determine how much grass seed they will need the caretaker measure the length and br eadth of the area of the sports field . He measured the length to be 162 m and the breadth 160 m. Calculate the area where the grass will be planted.
You may use the following formula:

Area of a rectangle = length x breadth (2)

3.1.2 Calculate the perimeter of the sports field in metres.
You may use the following formula:

Perimeter = 2 (length + breadth) (2)

3.1.3 Determine the number of holes the men will dig on the length sides only if there is one hole in every 2 m. There is a gate of 4 m that will be attached to two poles.

You may use the formula:

No. of holes = s m of en th sides
– 1 (3)

3.1.4 Calculate the surface area of the classroom that has to be painted. The area of the door is 1,71 m2 and the area for the windows is 4,875 m2.
You may use the following formula:

Surface area of classroom = 2(length x height) + 2(breadth x height) – (area of windows and door) (3)

3.1.5 The paint they chose indicates a spread rate of 6 m for every 1 litre of paint. The paint is sold in tins of 5 litres only. Calculate how many tins of 5 litres of paint they will need to buy. (3)

2

3.2 Rondo, a 15 -year-old boy from Nadia High School uses the bathroom scale as shown below to measure his mass in kilograms.

3.2 Determine Rondo's Body Mass Index (BMI) if he is 1 832 mm tall.
Use the formula:

BMI = mass in
hei ht in metres
(4)
[17]

QUESTION 4

4.1 On ANNEXURE B is the map of an extract of the eastern part of South Africa. Use this map to answer the following questions .

4.1.1 Write down the grid reference for Richards Bay. (2)

4.1.2 Identify the ocean that can be seen on the map. (2)

4.1.3 Two neighbo uring countries can be found north of Kwazulu -Natal. Write down the name of these countries. (2)

4.1.4 The distance, on the map, between Ladysmith (A2) and KwaDukuza (B2) is 6,5 cm. Use the map scale and calculate the actual distance (in kilometres) between these two towns. (3)

4.1.5 Draw the route from Phinda (C3) to Big Bend (B3). Indicate the town s on the route. (3)

4.1.6 Determine which province represents the largest portion of the map. (2)

4.1.7 Identify the two places south of Mount Ayliff on the map that you will pass before getting to Mount Ayliff (A1). (2)
[16]

QUESTION 5

The maternity sections of South African hospitals were a hive of activity with a total of 320 babies delivered on New Year's Day. (01/01/2014 at 19:30 SABC1 TV news). The data has been recorded in the table below:

TABLE 2: Table representing babies born on 1 January 2014
Provinces
Eastern
Cape
Limpopo
Mpumalanga
Northern
Cape
Western
Cape
Gauteng
Kwazulu -
Natal

No. of new -born

babies 43 105 25 18 36 60 33

5.1 5.1.1 Identify the province with the highest number of new-born babies on 1 January 2014. (2)

5.1.2 Express the number of babies born in Gauteng as a ratio to the total number of babies born on New Year's Day . Write the ratio in its simplest form. (4)

5.1.3 Complete the bar graph on the ANSWER SHEET using the above information for the missing provinces. (4)

5.2 In April 2012 the Department of Transport implemented e -tolls on some of Gauteng's major highways. The following is the fees charged by some of the gantries for Class A2 vehicles when passing underneath them.   Use TABLE 3 below to answer the questions below :

TABLE 3: Table depicting e -toll fees for some gantries in Gauteng.

Plaza Name and Place	Tariff for registered e -tag user Class A2	Tariff for a non -registered user Class A2
Babet (N1 -21)	R3,00	R5,80
Indlazi (N1 -21)	R2,91	R5,63
Flamango (N1 -21)	R2,76	R5,34
Mossie (N1 -21)	R3,00	R5,80
Tarentaal (N1 -20)	R2,58	R4,99
Owl (N1 -20)	R3,21	R6,21
Stork (N1 -20)	R2,52	R4,87
Kiewiet (N3 -12)	R2,31	R4,47
Sunbird (N1 -20)	R3,36	R6,50
Pelican (N1 -20)	R3,21	R6,21
Starling (N3 -12)	R2,46	R4,76
Phakwe (N12 -18)	R2,22	R4,29
Thaha (N12 -18)	R3,15	R6,09
Hadeda (R21 -1)	R2,43	R4,70
Gull (N12 -19)	R3,30	R6,38
Bee-eater (N12 -19)	R2,43	R4,70
Ihobe (N1 -21)	R3,36	R6,50
Blouvalk (N1-20)	R2,58	R4,99
Leeba (N3 -12)	R2,16	R4,18
Ivusi (N1 -21)	R3,36	R6,50

[Source: Government Gazette Volume 562 ; Pretoria, 13 April 2012; Issue No. 35263]

5.2.1 Calculate the mean tariff for non -registered e -tag user Class A2. (3)

5.2.2 Determine the median tariff for registered e -tag user Class A2. (3)

5.2.3 Determine the modal (mode) tariff for non -registered e -tag user Class A2. (2)

5.2.4 Determine the probability that a registered tariff amount that is randomly selected will be R3, 21. Write your answer in its simplified form. (2)

5.2.5 Determine the range for registered e -tag user tariff amounts . (2)

5.2.6 Identify the tariff amount charged for motorbikes . Use ANNEXURE C . (2)

[24]

TOTAL : 100

ANNEXURE A

QUESTION 1.1

QUOTATION

CUSTOMER INFORMATION:
Name: Johnson Louw
Industrial Contractors
Missionville
Jacksonville
1112 23 Mara Street
Collondale
Jacksonville
1112
Reg No.: 2001/023115/10
VAT No.: 8754896368
Tel No.: (043) 741 2020
Fax No.: (043) 741 6665
E-mail: Info@riohardware.co.za

Valid until ………. ……………. Quotation No.: Q000002158
DATE: 28/12/2013
SKU Description Quantity Unit Price
excluding VAT Total
excluding
VAT
FG00012 Rake double
sided
3
R   70,17
R   210,51
RBC -5622 Brush cutter
Ryobi 43cc
2
R2 367,54
R4 735,08
GRD1234 Bag refuse
plastic H/D 20*
3
R   28,06
R   84,18
HGF3457 Oil L/Star two
stroke 200 m■
1
R   28,94
R   28,94

TOTAL NET
VALUE
A

VAT INCLUDED
IN QUOTE   R
708,22
Thank you for your enquiry.
Please note that prices are subject to change and availability and may change without notice.
Delivery is excluded.
Standa rd terms and conditions apply.
BANKING DETAILS :

LLT Bank
Current Account number 7501213633
Branch code: 217 -213

RIO
HARDWARE
STORES

ANNEXURE B: MAP OF … PROVINCE IN SOUTH AFRICA

ANNEXURE C

Image of an e -toll gantry on the N1 freeway in Gauteng

Information of the different e -toll tariffs

Image of an e -tag

- A gantry is a metal frame that is fitted with cameras that record the cars that pass under it.
- An e-tag is a device that is fitted to your vehicle and it is electronically recording all the gantries that you pass under. The cost is then charged to your bank account linked to that device.

ANSWER SHEET

NAME: GRADE:

QUESTION 5.1.3

Provinces
Eastern Cape
Limpopo
Mpumalanga
Northern Cape
Western Cape
Gauteng
Kwazulu - Natal
No. of new-born babies 43 105 25 18 36 60 33

01020304050607080
Eastern Cape Limpopo Mpumalanga Nothern Cape Western Cape Gauteng KwaZulu NatalNumber of births
ProvincesNumber of births as on 1 January 2014

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2014

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages including a 1 page annexure .

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions. QUESTION 3.1 must be answered with reference to ANNEXURE A.

2. Number the questions correctly according to the numbering system used in this question paper.

3. An approved calculator (non-programmable and non-graphical) may be used, unless stated otherwise.

4. ALL calculations and steps must be shown clearly.

5. ALL the final answers must be rounded off to TWO decimal places, unless stated otherwise.

6. Start EACH question on a NEW page.

7. Write neatly and legibly.

QUESTION 1

Zeederberg High School has two Grade 12 classes, namely Grade 12A and Grade 12 B. These two classes are in competition with each other. This competition is about raising funds for the school.

1.1 Both of these classes will be selling cold drinks during break time. They buy the cold drinks in 2-litre bottles.

Grade 12 A uses 125 millilitre cups and sells it at R2, 00 each.
Grade 12 B uses 250 millilitre cups and sells it at R3, 50 each.

1.1.1 How many cups can Grade 12 A sell from ONE 2-litre bottle? (3)

1.1.2 How many cups can Grade 12 B sell from ONE 2-litre bottle? (3)

1.1.3 They bought the 2-liter cold drinks for R 8,99 each. From selling ONE 2-litre, Grade 12 B claims their profit will be greater than that of Grade 12 A, because their cups are bigger and they charge more per cup. Show with the necessary calculations whether you agree or disagree with Grade 12 B. (8)

1.1.4 Calculate the percentage profit that Grade 12 A made on selling ONE 2 litre-bottle.
Use the formula :

Percentage profit =
x 100

Give your final answer to the nearest 10 percent. (3)

1.2 The following stack bar graph shows the sales of the cold drinks of the two classes for two weeks. Use the graph to answer the questions below.

1.2.1 On which day or days did both classes sell the same amount of cups of cold drinks? (2)

1.2.2 Calculate the difference between the most number of cups of cold drinks and the least number of cups cold drinks sold per day for Grade 12 B. (3)

1.2.3 There was only ONE day that Grade 12 B sold more cups of cold drinks than Grade 12 A. Which day was it? (2)

1.2.4 What is the modal value of the number of cups of cold drinks for the two -week period for both classes? (2)

1.2.5 Grade 12 A claims that on average they sell more cold drinks than Grade 12 B. Do you agree , or disagree with this statement? Show all necessary calculations to justify your answer. (8)

1.2.6 Write the number of cups of cold drinks for Day 1 of Grade 12A and the number of cups of cold drinks for Grade 12B as a ratio in its simplest form. (2)
[36]

QUESTION 2

2.1 Vuyo bought tuna in bulk from a factory shop at a very good price. He wants to send some of it to his sister. Look at the following drawings (not drawn to scale) and answer the questions that follow.

Dimensions of the box Dimensions of the tuna tin
Length (l) = 48 cm
Width (w) = 35 cm
Height (h) = 15 cm Radius (r) = 4,2 cm
Height (h) = 3,8 cm

2.1.1 How many tuna tins can Vuyo pack across the length of the box? (4)

2.1.2 How many tuna tins can Vuyo pack across the width of the box? (3)

2.1.3 How many tuna tins can Vuyo stack on top of each other? (3)

2.1.4 Vuyo claims that he can get more than 50 tuna tins packed in the box. Show with the necessary calculations whether his statement is true. (3)

2.2 Vuyo has two options to get this box to his sister. He can either deliver it himself or he can make use of the Post Office Services.

■ The distance between Vuyo and his sister is 100 km. Vuyo's car uses 7,6 litres of petrol for every 100 km. At the time that Vuyo wanted to go to his sister, the petrol cost was R13,20 per litre.

■ The Post Office charges R 35,10 for the first kilogram and for each additional kilogram or a part of a kilogram R 4,70. The mass of the tuna tin is 170 gram.

Use the above information and determine by means of calculations which of these two options will be the most economical one for Vuyo. (8)
[21]

TUNA
h
w

l
h

r

QUESTION 3

3.1 Study the map of the Johannesburg City Centre, ANNEXURE A , and answer the questions below.

3.1.1 The traffic in Catherine Road (Grid B2) can only flow in one direction. What is this direction of flow? (2)

3.1.2 What do we call streets that only flow in one direction? (2)

3.1.3 Name any street where the traffic can only flow in an easterly direction. (2)

3.1.4 Give the grid reference of the Johannesburg College of Education. (2)

3.1.5 Marcia wanted to visit the Johannesburg Art Gallery (Grid C3). The person who directed her gave her the wrong directions and she found herself at the City Hall (Grid B3). Give Marcia detailed directions how to walk from the City Hall to the Johannesburg Art Gallery. (3)

3.1.6 Write the scale of the map as 1 : … Show all calculations. (3)

3.1.7 Marcia and her friend, Zoe, came from different directions to the Johannesburg Station (Grid B3). Marcia travels from the City Hall and Zoe from the Civic Centre (Grid B2). With the necessary calculations, who will be walking the shortest distance in metres? (6)

HINT : Use the centre of the dots as your starting points and
follow the roads.

3.2 While Marcia and Zoe are waiting on the train, they played a game with a pack of 52 playing cards. They are playing a game , called 7 cards where each one is dealt seven cards at the beginning of the game. They then have to build seven cards of the same suite. A pack of playing cards has 4 suites namely diamonds, hearts, clubs and spades.

3.2.1 How many of each suite are there before dealing the cards? (2)

3.2.2 What is the probability that the first card dealt to Zoe will be black and 10? (3)
[25]

QUESTION 4

4.1 For the type of work that Blythe is doing, he must have a reliable laptop or computer system. His laptop crashed and he is desperately in need of a new one. He saw the following advertisement in the newspaper. Study the advertisement carefully and answer the questions below.

4.1.1 Calculate the price of the package before the discount. (2)

4.1.2 The discount in the advert is R 900. Calculate this discount as a percentage to 1 decimal place. (3)

4.1.3 The fine print in the bottom right hand corner indicates “Interest rate from 8,5%. Total repayment from R 6 202,39.” Briefly explain the meaning of this fine print. (2)

4.1.4 If the discounted price is used with the interest rate of 8,5% per annum as indicated in the fine print, calculate the final amount. Also state whether the final value is the same as in the fine print and if not , give an explanation. Use the simple interest formula:

$A = P(1 + ni)$, where
A = Final amount
P = Original amount
i = interest rate
n = term in years (5)

4.1.5 Explain ONE disadvantage of buying an item on credit. (2)

4.2 The following pie chart shows what Blythe uses his laptop for. Answer the questions that follow.

4.2.1 Give a suitable heading for the pie chart. (2)

4.2.2 Name any TWO usages of the laptop that fall under the category of “other”. (2)

[18]

TOTAL: 100

QUESTION 3.1

ANNEXURE A

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2013

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2 hours

This question paper consists of 9 pages.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.

2. Answer ALL the questions .

3. Number your answers correctly according to the numbering system used in the question paper.

4. A non -programmable and non-graphical calculator may be used, unless stated otherwise.

5. ALL calculations and steps must be shown clearly.

6. ALL final answers must be rounded off to TWO decimal places, unless stated otherwise.

7. Units of measurement must be indicated where applicable .

8. Start EACH question on a NEW page.

9. Write neatly and legibly.

QUESTION 1

Linda and Thabo have decided to start up their own business baking and selling fresh home-made giant muffins, from their own home. This way they cut costs, as they do not have to pay rent for their business. To make sure that their business will make a profit they did a very careful costing of all their expenses before starting up and determining a selling price for their giant muffins.

Below is Table 1 showing their fixed monthly costs for their business. These are costs that remain the same no matter how much they produce and sell.

Fixed Monthly Expenses (R)
Salaries (for both of them) R6 000,00
Wages (1) R1 500,00
Water R465,00
Cleaning materials R250,00
Transport costs R1 750,00
Miscellaneous costs R500,00

1.1 Calculate the cost of the Fixed Monthly Expenses for their business. Money also needs to be spent on ingredients to make the giant muffins. (3)

Below in Table 2 showing the cost of ingredients needed to make 50 giant muffins.

Ingredient Costs for 50 Giant Muffins (R)
Flour R15,50
Baking Powder R0,25
Sugar R11,50
Oil R9,50
Eggs R12,75
Paper cake cups R8,00
Cocoa, Choc chips, Blueberries, Nuts, Flavours R12,50

The stove they use is an electric stove and by watching the meter they have worked out that they use 6 units (kWh) of electricity to bake 50 giant muffins. One unit (kWh) of electricity costs R1,09.

1.2 Calculate the cost for the ingredients to make 50 giant muffins. (3)

1.3 Calculate the cost of ingredients for ONE giant muffin. (2)

1.4 Calculate the cost of electricity to make ONE giant muffin. Give your answer in Rand and round your answer off to 2 decimal places. (2)

1.5 How much does it cost to make ONE giant muffin? (2)

1.6 Linda and Thabo have decided to sell their giant muffins at R5,00 each.

1.6.1 How much profit is made per giant muffin?
Use the following method to calculate the profit:

Profit = Selling Price – Cost price (3)

1.6.2 Calculate the percentage profit per giant muffin. Use the following formula to calculate your answer :

(3)

1.7 How many giant muffins must Linda and Thabo sell to break even all their costs and cover ?

Please round off your answer to the nearest whole muffin.
Use the following formula for your calculations :

(3)

1.8 Linda and Thabo had to have their kitchen altered to allow them to bake the number of muffins required for their business and to meet the standards required by the Health Department to allow them to run their business from home. It is worth doing this as it would add value to their property. The amount quoted to do the alterations was R11 505, 35. They decided to go to the bank to take a loan to cover these costs. They decided to borrow R12 500,00 in case there were some hidden costs. The bank was prepared to lend them money at 12,5% interest per annum over a period of 5 years using simple interest.

1.8.1 Calculate how much they would pay back to the bank after 5 years.

Use the formula :

Where = Final Amount
= The initial amount borrowed
= interest per annum
= number of years (3)

1.8.2 Calculate the monthly payment on this loan. (3)

1.8.3 Calculate how many extra muffins will need to be baked to have enough profit to cover this monthly payment.
Give your answer to the nearest whole muffin. (3)
[30]

QUESTION 2

The Smiths recently had a pool build in their back garden. The pool is 15 metres long and 8 metres wide. The depth of the pool is 1, 5 metres. A diagram of the pool is given below.

15 m
8 m

1,5 m

2.1 The pool walls need to be tiled to keep maintenance to a minimum.

2.1.1 Calculate the surface area of the walls of the pool. Give your answer in

Use the following formula to do your calculations:

(3)

2.1.2 If the tiles used measure 20 cm x 20 cm , calculate how many tiles will be needed to tile the pool walls. (4)

2.2 Calculate how much water is needed to fill the pool. Give your answer in kilolitres.

Use the following formula: Volume = Length x Width x Depth (3)

2.3 Water costs R5, 55 per kilolitre. How much will it cost to fill this pool? (3)

2.4 Last week the maximum temperature reached 39,5 °C and 1, 5% of the pool water evaporated. How much water evaporated out of the pool that day? (3)

2.5 The Smiths have decided to put paving stones around their pool to stop too much grass getting into their pool. The diagram on the next page shows the final measurements required.

17 m

15 m

10 m 8 m POOL

PAVING

2.5.1 What is the width of the paving required? (2)

2.5.2 If the paving stones used measure 50 cm x 50 cm, how many paving stones will be needed to complete the paving? (4)
[22]

QUESTION 3

Your youngest little sister, Lisa, is going to Grade 1 next year. Lisa is afraid of getting lost around her new school. She likes to know exactly where she needs to go at all times , or else she tends to panic and cry. As she is going to your old primary school you are able to help make getting around the school easier for her. You draw a sketch of the school as shown below.

N

Ms. Tshoba Ms Smith Ms Faith MsTabata
Hall Grade 1A Grade 1B Grade 3B Grade
4 A

Veranda Principal’s
Girls’ toilet s Grass Office

Boys’ toilets Secretar y
Office
Mr Naidoo Ms Time Ms Wilson Ms Styles
Library
Grade 4B Grade 2A Grade 2B Grade
3A Computers

Scale = 1 : 400

3.1 All learners will enter the school through the front door on the first day. No learners are allowed on the grass except at break time and under supervision of a teacher.

3.1.1 Your sister is going to Grade 1. On which side of the school is Lisa’s class? Use a compass direction for your answer. (1)

3.1.2 Lisa is going into Ms Smith’s class next year. Give her directions how to get from the front door of the school to Ms Smith’s classroom. (2)

3.1.3 Is there a single or double door going out to the playing fields? Why do you think this is so? (2)

3.2 Lisa want s to know how big her classroom is.

3.2.1 Measure the length and breadth of her classroom on the drawing. Give your answer in centimetres. (2)

3.2.2 Using the scale provided, give the real life length and breadth of her classroom. (4) Playing Fields
Front of School

3.3 Draw the front v iew of the school building. Remember to show all windows, doors and the roof . (3)

3.4 Are all the classrooms in the school the same size? Give a reason why you think this is so. (2)

3.5 Lisa’s school fees are R 950 per month for ten months.

3.5.1 Calculate the annual cost of the school fees. (2)

3.5.2 If school fees are paid in full at the beginning of the year, 15% discount is given. How much discount is this? (2)
[20]

QUESTION 4

4.1 Earlier this year South Africa played against New Zealand in a T20 series at Newlands Cape Town.

Below is a graph showing the number of runs made per over for both teams. Study the graph carefully and answer the questions that follow.

4.1.1 In which over did the Proteas score the most runs? (1)

4.1.2 In which over did the Proteas score no runs at all? (1)

4.1.3 In how many overs did the Proteas score 5 runs or less? (2)

4.1.4 In how many overs did the New Zealanders score 12 runs each? (2)

4.1.5 During this match, what was the probability that the batsman would hit 9 runs in an over for South Africa? (2) 051015202530
1 2 3 4 5 6 7 8 910 11 12 13 14 15 16 17 18 19 20Number of Runs
Overs Proteas vs New Zealand T20
Proteas
New Zealand

4.1.6 What is the total number of runs hit by South Africa for this match? (2)

4.1.7 Calculate the average number of runs per over for South Africa. (2)

4.1.8 Which team won this match? Show all your calculations to support your answer. (3)

4.2 Recently a survey was done amongst 400 learners at school to find out what cellphones get used the most for by teenagers.

Below is a pie chart showing the results of the survey.

4.2.1 According to the survey, what are the cell phones used for most ? (1)

4.2.2 What percentage of learners uses their cell phones for taking photos? Use the following method to calculate the percentage :

x 100 (3)

4.2.3 What do 8% of learners use their cell phones for? (3)

4.2.4 Calculate the number of degrees that were used to draw the sector for Social Networking . (3)

4.2.5 If any of the surveyed learners were asked , what would be the probability that the learners use their cell phone for games? Write your answer as a percentage. Show all calculations used. (3)
[28]

TOTAL: 100
SMS, 32
INTERNET, 33
MUSIC, 61
CALLS, 41
SOCIAL
NETWORK, 80 GAMES, 45 PLEASE CALL ME,
15 PHOTOS, 72 RECORDINGS, 21 CELLPHONE USAGE BY LEARNERS

Province of the
EASTERN CAPE
EDUCATION

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2012

MATHEMATICAL LITERACY P1

MARKS: 100

TIME: 2½ hours

This question paper consists of 8 pages , including an annexure .

MLITE1

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.

2. Answer ALL the questions .

3. QUESTIONS 4.2.2 and 4.2.3 must be answered on the attached annexure . Write your name in the space provided on the annexure and hand it in with your ANSWER BOOK.

4. Number your answers correctly according to the numbering system used in the question paper.

5. A non -programmable and non -graphical calculator may be used, unless stated otherwise.

6. ALL calculations and steps must be shown clearly.

7. ALL final answers must be rounded off to TWO decimal places, unless stated otherwise.

8. Units of measurement must be indicated where applicable .

9. Start EACH question on a NEW page.

10. Write neatly and legibly.

(NOVEMBER 2012) MATHEMATICAL LITERACY P1 3

QUESTION 1

1.1 DVD Musika is having a sale. All CDs are reduced by 15% and all DVDs are marked down by a . Calculate the following:

1.1.1 If CDs were marked at R125 ,00, how much will you pay for them while they a re on sale? (2)

1.1.2 If all DVDs were R215, calculate what their sale price will be. (3)

1.2 During the holidays Peter spent 2 hours gardening. He spent 45 minutes mowing the lawn, 20 minutes trimming the edges, 30 minutes weeding and the remainder of the time watering the flowers. What fraction of the time (in the simplest form) did Peter spend:

1.2.1 mowing the lawn? (2)

1.2.2 watering the flowers? (4)

1.3 Naledi has been asked to bake 50 cheese muffins for a school function. After looking in all her recipe books she found the following recipe:

Cheese Muffins
(makes 30)

600 g Flour
50 m ■ Baking Powder
450 g Grated Cheese
60 m ■ Oil
750 m ■ Milk

To assist Naledi, calculate how much she needs of the following to make the 50 cheese muffins required:

1.3.1 Flour (2)

1.3.2 Grated Cheese (2)

1.4 Peter and Naledi went to George with their parents during the holidays. It took them 6 hours to travel 561 km from King William’s Town to George. The car’s petrol tank was full when they left King William’s Town. When they stopped at George the tank was filled up with 31,17 litres of petrol.

1.4.1 What was their average speed for the trip? Give your answer to the nearest km .
Use the formula:
Distance travelled = speed x time (4)

1.4.2 Calculate how far they travelled on 1 litre of petrol. Give your answer correct to the nearest km. (3)

1.4.3 Calculate the cost of the petrol, if the petrol price was R10,05 when they filled up in George. (3)
[25]

QUESTION 2

Xolani is making and selling wooden coffee tables for a living. The graphs below show his expenses and income for a month.

Use the graphs to answer the following questions:

2.1 What are Xolani’s minimum fixed expenses even when he has made no coffee tables at all? Say what you think this cost is . (2)

2.2 Give TWO examples of expenses that Xolani has in making coffee tables. (2)

2.3 How many coffee tables must Xolani sell in order to break even? (2)

2.4 What are Xolani’s expenses when he makes 20 coffee tables? (2)

2.5 What is Xolani’s income when he sells 20 coffee tables? (2)

2.6 How much profit has Xolani made when he sells 20 coffee tables? (3)

2.7 Calculate how much Xolani sells 1 coffee table for. (3)

2.8 What will be Xolani’s income when he sells 15 coffee tables? (2)

2.9 If Xolani were to sell his coffee tables for R60 what would happen to his profit? Show the reason for your answer using calculations. (4)

2.10 If Xolani has an income of R585,00, how many tables has he sold? (2)
[24]
01002003004005006007008009001000
01234567891011121314151617181920Amount in Rands
Number of coffee tables Xolani's Coffee Table Business
Expenses
Income

QUESTION 3

Cereal is sold in boxes to prevent it from breaking while being transported and handled. The company has to be careful in the design of the box so as to not use too much cardboard. The picture below show the packaging decided upon and the measurements.

3.1 Draw a net for the cereal box showing all the necessary measurements. (4)

3.2 Calculate the surface area of the box in cm2. Remember there are six sides to the box and each side is a rectangular shape.
The area for a rectangle = length x breadth (5)

3.3 If cardboard cost 0,06 cents per cm2, calculate how much the cardboard for 1 box will cost. (3)

3.4 If printing of the box costs 0,04 cents per cm2, calculate the cost for printing 1 box. (3)

3.5 Calculate the volume of the box.
Use the formula: Volume = length x breadth x height (3)

3.6 If 7,5 g of cereal has a volume of 1 cm3, calculate how many grams of cereal will fit into this box. (3)

3.7 The cereal boxes get cut from rolls of card measuring 2 m by 15 m. The box nets are placed next to each other without any spaces in between. Calculate how many boxes can be cut from one roll of card. (5)
[26]
25 cm
7,5 cm 20 cm

QUESTION 4

4.1 A recent survey was done to find out which cars were the most popular in South Africa. The following two graphs show the results obtained in two different suburbs in Johannesburg .

4.1.1 Which is the most popular car in Suburb A? (1)

4.1.2 Which is the most popular car in Suburb B? (1)

4.1.3 What can you tell about the income of the residents of the two suburbs? (2)

4.1.4 If you were to use just the two surveys given, would they give you a true picture of the most popular car in South Africa? Give a reason for your answer. (2)

4.2 The Grade 11A class was given an English Language test out of 100 last week. These were the results in ranked order :

23 41 42 50 50 51 54 55 56 57
60 61 65 66 66 67 68 69 70 70
70 72 74 76 79 82 85 86 88

4.2.1 Calculate the following:

(a) The mean mark (3)

(b) The mode (1)

(c) The median (2)
010203040506070
Palio Citi
GolfBMW Audi TazzNumber of Cars
Make of Car Suburb A
0102030405060708090Number of Cars
Make of Car Suburb B

4.2.2 Complete the frequency table below using the table given on the annexure for your answers.

Interval	Frequency	
20 – 29		
30 – 39		
40 – 49		
50 – 59		
60 – 69		
70 – 79		
80 – 89		(4)

4.2.3 Draw a histogram using the results from your frequency table. Use the grid on the annexure to construct the histogram. (5)

4.2.4 Which measure of central tendency in QUESTION 4.2.1 a – c tells you the most about how the class performed in the test? Give a reason for your answer. (2)

4.2.5 Was this test too difficult or too easy? Motivate your answer. (2)
[25]

TOTAL: 100

ANNEXURE

NAME: GRADE:

QUESTION 4.2.2

Interval	Frequency	
20 – 29		
30 – 39		
40 – 49		
50 – 59		
60 – 69		
70 – 79		
80 – 89		(4)

QUESTION 4.2.3
ENGLISH TEST RESULTS

Frequency

20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89

Test Results
(5)